

Cervical Cancer Prediction Using Machine Learning

**A Project Report submitted in the partial fulfillment for the Requirements
for the award of the degree**

MASTER OF COMPUTER APPLICATIONS

Submitted by

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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

NARASARAOPETA ENGINEERING COLLEGE: NARASARAOPET

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2025-2026

Cervical Cancer Prediction using Machine Learning

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Plagiarism Report:

Submission date: Apr 27, 2026, 9:37 AM GMT+5:30

Submission ID: 3551783758

File name: 24471F0096_Document.pdf

Word count: 7,610

Character count: 44,082

Similarity Index: 43%

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CERTIFICATE

This is to certify that the project work entitled **“Cervical Cancer Prediction using Machine Learning”** is being submitted by Mr. Sikhakolanu Deepak Naga Sai (24471F0096) as a partial fulfillment of the requirements for the award of MASTER OF COMPUTER APPLICATIONS degree of JNTUK, Kakinada during December 2025 – April 2026 under my supervision and found worth of acceptance to requirements of the university.

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DECLARATION

I here by declare that this project work titled **“Cervical Cancer Prediction using Machine Learning”** is a bonafied work carried out by me under the guidance and supervision of Mrs.A. Aruna Kumari, Mca, Assistant Professor, Dept of MCA, Narasaraopeta Engineering College, for the award of Master of Computer Applications degree of Jawaharlal Nehru Technological University Kakinada. This work has not been submitted to any other University/Institution for the award of any degree.

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ACKNOWLEDGEMENT

I wish to express my heartfelt gratitude to all those who have supported and guided me throughout the successful completion of this work.

First and foremost, I am deeply grateful to our esteemed Chairman, **Sri M.V. Koteswara Rao**, B.Sc., whose unwavering encouragement and continuous support have been instrumental throughout my academic journey.

I would like to extend my sincere thanks to our respected Principal, **Dr. S. Venkateswarlu**, M.E., Ph.D., for his valuable guidance and encouragement, which have greatly contributed to my growth and learning.

My deepest appreciation goes to **Dr. K. Lakshmi Nadh**, M.Tech., Ph.D., Professor and Head of the MCA Department, whose insight and leadership have provided the foundation for academic excellence. I also express my sincere thanks to my project guide, **Mrs. M. Mounika Naga Bhavani**, M. Tech, Assistant Professor, for her patient guidance, invaluable suggestions, and constant motivation throughout the course of this project.

I am especially thankful to **Mr. G. Ravi Purna Chandra Rao**, MCA, Assistant Professor and Project Coordinator, for his support, constructive feedback, and encouragement, which have been a constant source of inspiration during this work.

I extend my appreciation to the faculty and staff of the MCA Department for their ongoing support, insightful discussions, and encouragement, which have enriched my learning experience..

Finally, I thank my friends and peers for their helpful suggestions, collaboration, and moral support, all of which have contributed meaningfully to the successful completion of this work.

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ABSTRACT

Cervical cancer is one of the most common and life-threatening cancers affecting women worldwide, especially in developing countries. Early detection of cervical cancer plays a crucial role in reducing mortality rates and improving the survival chances of patients. Traditional diagnostic methods such as Pap smear tests and biopsies are time-consuming, costly, and highly dependent on medical expertise. Hence, there is a need for an automated and accurate system to assist in early diagnosis.

This project, titled “Cervical Cancer Prediction Using Machine Learning”, aims to develop an efficient machine learning-based system to predict the occurrence of cervical cancer using medical and demographic data. The proposed system uses a publicly available cervical cancer dataset containing various risk factors such as age, number of pregnancies, sexual behavior, and medical history. The dataset is preprocessed to handle missing values and improve data quality.

Machine learning algorithms are trained on the processed dataset to classify whether a patient is likely to have cervical cancer or not. The performance of the system is evaluated using metrics such as accuracy, precision, recall, and confusion matrix. The experimental results show that machine learning techniques can provide reliable and accurate predictions, thereby assisting medical professionals in early diagnosis and decision-making.

The proposed system is cost-effective, efficient, and can be used as a supportive tool in healthcare environments. This project demonstrates the potential of machine learning in medical diagnosis and highlights its importance in improving healthcare services.

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1. INTRODUCTION

Cervical cancer is a type of cancer that develops in the cervix, which is the lower part of the uterus connecting to the vagina. It is one of the leading causes of cancer-related deaths among women globally [1]. Cervical cancer usually develops slowly and may not show symptoms in its early stages, making early detection difficult. If detected at an early stage, cervical cancer can be treated effectively and survival rates can be significantly improved.

In recent years, advancements in medical technology and data analysis have enabled the use of computational techniques for disease prediction [2]. Machine learning is a branch of artificial intelligence that allows systems to learn patterns from data and make predictions without being explicitly programmed [3]. The application of machine learning in healthcare has gained significant attention due to its ability to analyze large and complex datasets efficiently [4].

This project focuses on applying machine learning techniques to predict cervical cancer by analyzing patient risk factors. By using machine learning models, the system can assist doctors in identifying high-risk patients at an early stage.

Traditional methods for diagnosing cervical cancer involve clinical tests such as Pap smear tests, HPV tests, and biopsies [5]. These methods require specialized equipment, skilled professionals, and considerable time for diagnosis. In many cases, delayed diagnosis leads to the progression of the disease, reducing the chances of successful treatment.

Additionally, manual diagnosis may lead to human errors due to fatigue or misinterpretation of test results. There is a need for an automated system that can analyze patient data efficiently and provide accurate predictions to support medical professionals [6].

The problem addressed in this project is to design and implement a machine learning-based system that can predict cervical cancer using patient data, thereby reducing diagnostic time, cost, and dependency on manual analysis.

1.1 Objectives of the Project

The main objectives of this project are:

- To study and analyze cervical cancer risk factors
- To design a machine learning-based prediction system
- To preprocess and analyze the cervical cancer dataset

- To train and evaluate machine learning models for prediction
- To improve early detection and decision-making in healthcare

1.2 Scope of the Project

The scope of this project is limited to predicting the presence of cervical cancer using historical medical and demographic data. The system acts as a decision-support tool and does not replace medical diagnosis. The project demonstrates how machine learning techniques can be effectively applied in the healthcare domain to assist doctors and medical practitioners.

Future enhancements may include the integration of deep learning models, real-time hospital data, and deployment as a web or mobile application for broader accessibility.

Overview of Cervical Cancer Prediction Using Machine Learning

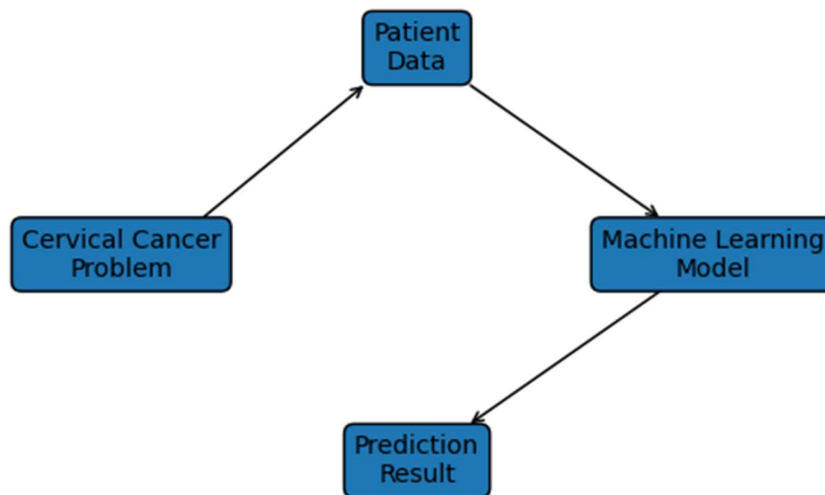


Figure 1.1 Overview of Cervical Cancer Prediction Using Machine Learning

Figure 1.1 The diagram titled “Overview of Cervical Cancer Prediction Using Machine Learning” illustrates the basic workflow of predicting cervical cancer using machine learning techniques. It presents a simple step-by-step process that begins with identifying the cervical cancer problem and ends with generating a prediction result.

The first component in the diagram is Cervical Cancer Problem, which represents the medical challenge of detecting cervical cancer at an early stage. Early detection is very important because it helps doctors provide timely treatment and improves patient survival rates.

Next, the process moves to Patient Data. This includes medical information collected from patients such as age, medical history, HPV infection status, lifestyle factors, and results from medical tests. These data points are essential for training and evaluating machine learning models.

The collected patient data is then provided to the Machine Learning Model. In this stage, algorithms analyze patterns within the data and learn relationships between different medical factors and the occurrence of cervical cancer. Common algorithms may include decision trees, random forests, or logistic regression.

Finally, the model produces a Prediction Result, which indicates whether a patient is likely to have cervical cancer or not. This result can assist healthcare professionals in making better diagnostic decisions and improving early screening programs.

2. LITERATURE SURVEY

Title 1: CerviFormer: Pap-Smear Based Cervical Cancer Classification

Author(s): Bhaswati Singha Deo et al.

Year: 2023

Summary: This paper presents a transformer-based deep learning model called CerviFormer for cervical cancer detection using Pap smear images [7]. The model uses attention mechanisms to capture complex features from medical images. It improves classification performance compared to traditional CNN models. The approach achieves high accuracy and demonstrates the effectiveness of transformer architectures in healthcare.

Advantages:

- High accuracy with deep learning.
- Works well on image data.
- Advanced transformer architecture.

Disadvantages:

- Requires large dataset.
- High computational cost.

Title 2: Cervical Cancer Prediction Using Machine Learning

Author(s): Rashmi Ashtagi et al.

Year: 2024

Summary: This study applies machine learning algorithms such as Random Forest, SVM, and Logistic Regression for cervical cancer prediction [8]. It uses patient risk factors like age, smoking, and medical history. The results show that ensemble methods provide better accuracy compared to single models. The study highlights the importance of feature selection in improving performance.

Advantages:

- Uses real-world clinical data.
- Good accuracy with multiple models.

Disadvantages:

- Limited dataset size.
- Feature dependency.

Title 3: Cervical Cancer Prediction Based on Imbalanced Data

Author(s): M.M. Muraru et al.

Year: 2024

Summary: This paper focuses on handling imbalanced datasets in cervical cancer prediction [9]. It uses SMOTE and other resampling techniques to balance the data. The study improves classification performance by reducing bias toward majority classes. It also evaluates different machine learning models under balanced conditions.

Advantages:

- Handles imbalance problem.
- Improves accuracy.

Disadvantages:

- Risk of overfitting.
- Complex preprocessing.

Title 4: Comparative Study on ML Classifiers for Cervical Cancer

Author(s): Various.

Year: 2024

Summary: This research compares different machine learning algorithms for cervical cancer prediction [10]. Models such as SVM, Decision Tree, and Naive Bayes are evaluated. The results show that SVM achieves the highest accuracy among all models. The study provides insights into selecting suitable algorithms for prediction tasks.

Advantages:

- High accuracy models.
- Comparative analysis.

Disadvantages:

- Overfitting risk.
- Limited generalization.

Title 5: Interpretable AI for Cervical Cancer Prediction

Author(s): P. Roy et al.

Year: 2025

Summary: This paper introduces explainable AI techniques for cervical cancer prediction [11]. It uses models that provide feature importance and decision transparency. The approach helps doctors understand the reasoning behind predictions. It improves trust and usability of machine learning systems in healthcare.

Advantages:

- Transparent predictions.
- Better decision-making.

Disadvantages:

- Slightly lower accuracy.
- Increased complexity.

Title 6: Deep Learning Methods for Cervical Cancer Detection

Author(s): P. Patre et al.

Year: 2025

Summary: This study reviews deep learning methods such as CNN and transfer learning [12]. It explains how image-based detection improves early diagnosis of cervical cancer. Data augmentation techniques are used to enhance model performance. The paper highlights the importance of deep learning in medical imaging.

Advantages:

- High Performance.
- Suitable for image data.

Disadvantages:

- Needs large Dataset.
- High Training Time.

Title 7: ML-Based Prediction of Clinical Outcomes**Author(s):** G. Bai et al.**Year:** 2025

Summary: This research uses machine learning models to predict clinical outcomes in cervical cancer patients [13]. It analyzes patient data such as lab results and medical history. The model helps in predicting disease progression and survival rates. It supports doctors in making better treatment decisions.

Advantages:

- Useful for prognosis.
- Real-world application.

Disadvantages:

- Data Quality Issues.
- Limited Scalability.

Title 8: ML Models for Cervical Cancer Risk Prediction**Author(s):** J. Su et al.**Year:** 2025

Summary: This paper develops machine learning models for predicting cervical cancer risk [14]. It uses SHAP values to explain model predictions. The approach identifies important features influencing the results. It improves both accuracy and interpretability of the system.

Advantages:

- Explainable results.
- Feature importance analysis.

Disadvantages:

- Computationally expensive
- Requires expertise.

Title 9: Federated Learning for Cervical Cancer Prediction

Author(s): S. Singla et al.

Year: 2025

Summary: This study introduces federated learning for privacy-preserving cervical cancer prediction [15]. It allows multiple institutions to train models without sharing data. Blockchain technology is used to enhance security. The model achieves high accuracy while maintaining data privacy.

Advantages:

- Ensures privacy.
- High accuracy.

Disadvantages:

- Complex system
- Infrastructure requirement.

Title 10: AI-Based Prediction Using Imaging and Clinical Data

Author(s): M. Zhang et al.

Year: 2025

Summary: This paper combines imaging data and clinical information for cancer prediction. Machine learning models such as SVM are used for classification. The system improves prediction accuracy by using multiple data sources [16]. It demonstrates the benefits of multi-modal data integration.

Advantages:

- Better accuracy.
- Uses multiple data sources.

Disadvantages:

- Complex Design.
- High Computation.

3. SYSTEM ANALYSIS

3.1 Existing System

In the existing system, cervical cancer diagnosis is mainly performed using traditional medical tests such as Pap smear tests, HPV tests, colposcopy, and biopsy [17]. These methods require clinical infrastructure, laboratory facilities, and skilled medical professionals. The diagnosis process is time-consuming and often expensive, making it less accessible in rural and developing regions.

The existing system heavily depends on manual examination and expert interpretation of test results. This may lead to delayed diagnosis and, in some cases, human errors [18]. Additionally, the traditional methods do not efficiently utilize historical patient data for predictive analysis [19].

3.1.1 Disadvantages of Existing System

The disadvantages of the existing system are as follows:

- High cost of diagnostic procedures
- Time-consuming testing and analysis
- Dependency on skilled medical experts
- Possibility of human error
- Limited accessibility in remote areas
- Lack of automated prediction support

3.2 Proposed System

The proposed system is a machine learning-based cervical cancer prediction system that analyzes patient medical and demographic data to predict the likelihood of cervical cancer. The system uses historical data and applies machine learning algorithms to identify patterns and relationships among risk factors.

The proposed system includes data preprocessing, feature selection, model training, and prediction phases. By using machine learning techniques, the system can provide faster and more accurate predictions, supporting healthcare professionals in decision-making.

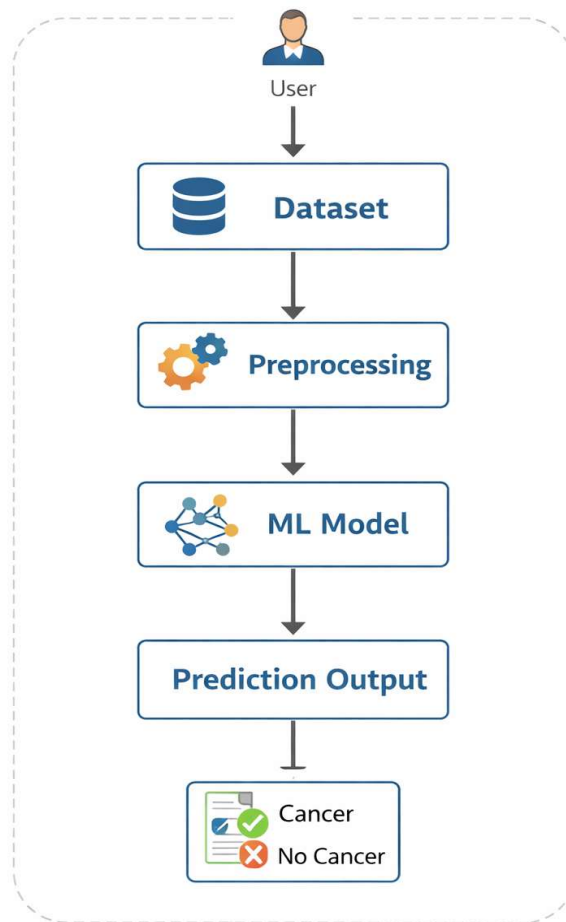


Figure 3.2.1 Machine Learning Workflow for Cervical Cancer Prediction

Figure 3.2.1 The diagram illustrates the workflow of predicting cervical cancer using a machine learning approach. The process begins with the User, who provides or accesses the Dataset containing patient medical information. This dataset then goes through a Preprocessing stage where missing values are handled, irrelevant data is removed, and the data is prepared for analysis. After preprocessing, the cleaned data is fed into the Machine Learning (ML) Model, which learns patterns and relationships from the data. Finally, the system generates a Prediction Output, indicating whether the patient is likely to have Cancer or No Cancer, helping support early detection and medical decision-making.

3.2.2 Advantages of Proposed System

The advantages of the proposed system include:

- Early detection of cervical cancer
- High prediction accuracy
- Reduced diagnosis time
- Cost-effective solution
- Minimizes human error
- Can handle large datasets efficiently
- Acts as a decision support tool for doctors

3.3 Feasibility Study

Feasibility study is conducted to evaluate whether the proposed system is practical and viable for implementation. The feasibility analysis is divided into three categories: technical feasibility, economic feasibility, and social feasibility.

3.3.1 Technical Feasibility

The proposed system is technically feasible as it uses widely available technologies such as Python programming language and machine learning libraries. The implementation is carried out using Google Colab, which provides a free and cloud-based development environment. Machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are well-supported by existing libraries.

3.3.2 Economic Feasibility

The system is economically feasible as it uses open-source tools and technologies. Python, Google Colab, and machine learning libraries are available free of cost. No additional hardware investment is required, making the system cost-effective and affordable.

3.3.3 Social Feasibility

The proposed system is socially feasible as it aims to improve women's healthcare by enabling early detection of cervical cancer. The system can assist medical professionals and improve healthcare services, especially in underdeveloped regions. It also raises awareness about the importance of preventive healthcare.

3.4 System Flow Diagram

The system follows the below flow:

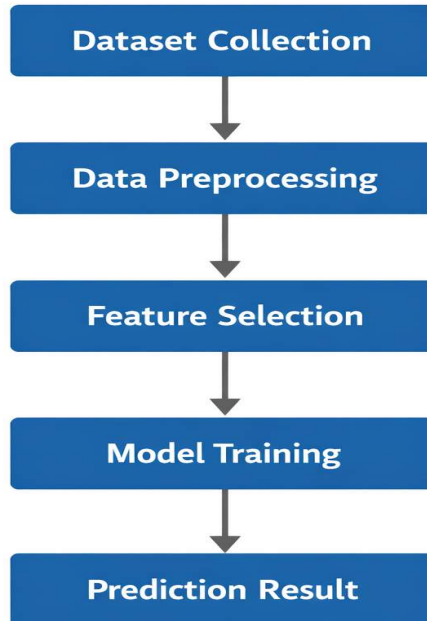


Figure 3.4.1

Figure 3.4.1 The system flow diagram illustrates the overall working process of the cervical cancer prediction system. Initially, the dataset is collected and loaded into the system for analysis. The data preprocessing step cleans the data by handling missing and inconsistent values. Feature selection is then performed to identify the most important attributes influencing cervical cancer prediction.

4. SYSTEM REQUIREMENTS

4.1 Software Requirements

The following software tools are required to develop and implement the proposed system:

- **Operating System:** Windows / Linux
- **Programming Language:** Python
- **Development Environment:** Google Colab
- **Libraries:**
 - NumPy
 - Pandas
 - Matplotlib
 - Scikit-learn

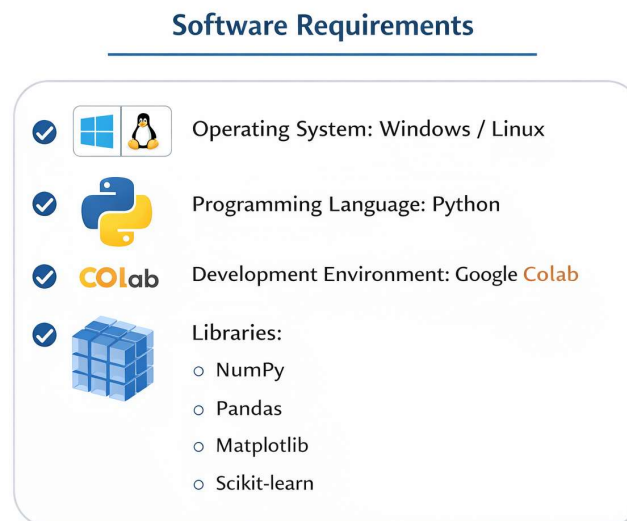


Figure 4.1.1 Software Requirements for Cervical Cancer Prediction System

Figure 4.1.1 The diagram presents the software requirements needed to develop a machine learning system for cervical cancer prediction. It specifies the Operating System as either Windows or Linux, which provides the platform to run the application. The Programming Language used is Python, known for its strong support for data analysis and machine learning tasks. The Development Environment mentioned is Google Colab, which allows users to write and execute Python code in a cloud-based environment. The system also uses several important

libraries, including NumPy for numerical operations, Pandas for data handling, Matplotlib for visualization, and Scikit-learn for implementing machine learning algorithms.

Python is chosen due to its simplicity, flexibility, and wide support for machine learning libraries. Google Colab provides a cloud-based platform with pre-installed libraries, making development faster and easier.

4.2 Hardware Requirements

The hardware requirements for the proposed system are minimal:

- **Processor:** Intel i3/i5 or higher
- **RAM:** Minimum 4 GB (8GB recommended)
- **Storage:** 20 GB free disk space
- **System Type:** 64-bit Operating System
- **Internet Connection:** Required for cloud-based execution

4.3 Functional Requirements

Functional requirements describe what the system is expected to do.

- The system should load the cervical cancer dataset
- The system should preprocess the dataset
- The system should train machine learning models
- The system should predict cervical cancer occurrence
- The system should display prediction results

4.4 Non-Functional Requirements

Non-functional requirements describe how the system performs.

- **Accuracy:** High prediction accuracy
- **Performance:** Fast execution
- **Reliability:** Consistent results
- **Scalability:** Ability to handle large datasets
- **Usability:** Easy to understand and use

4.5 Software Description

4.5.1 Machine Learning

Machine learning is a subset of artificial intelligence that enables systems to learn from data and make predictions without explicit programming. Machine learning algorithms analyze historical data, identify patterns, and use these patterns to predict future outcomes. Machine learning plays a significant role in healthcare by assisting in disease diagnosis and prediction.

4.5.2 Types of Machine Learning

Machine learning can be classified into the following types:

- **Supervised_Learning:**

Uses labeled data to train models. This project uses supervised learning.

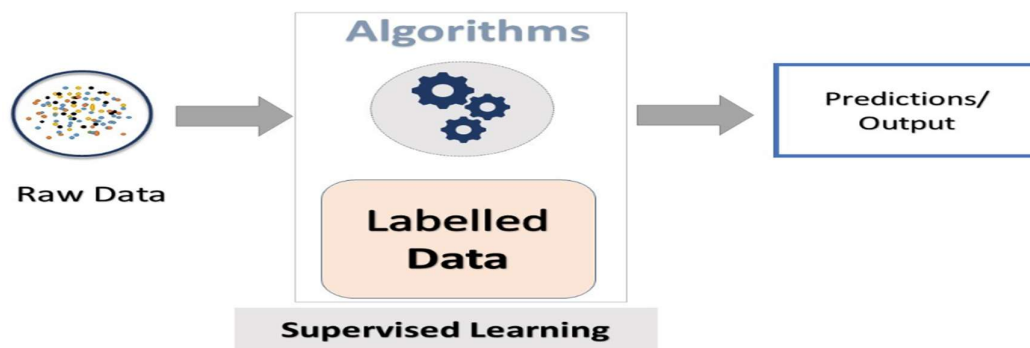


Figure 4.5.2.1 Supervised Learning Diagram

- **Unsupervised_Learning:**

Works on unlabelled data to find hidden patterns.



Figure 4.5.2.2 Unsupervised Learning Diagram

- **Reinforcement Learning :**

Learns by interacting with the environment using rewards and penalties.

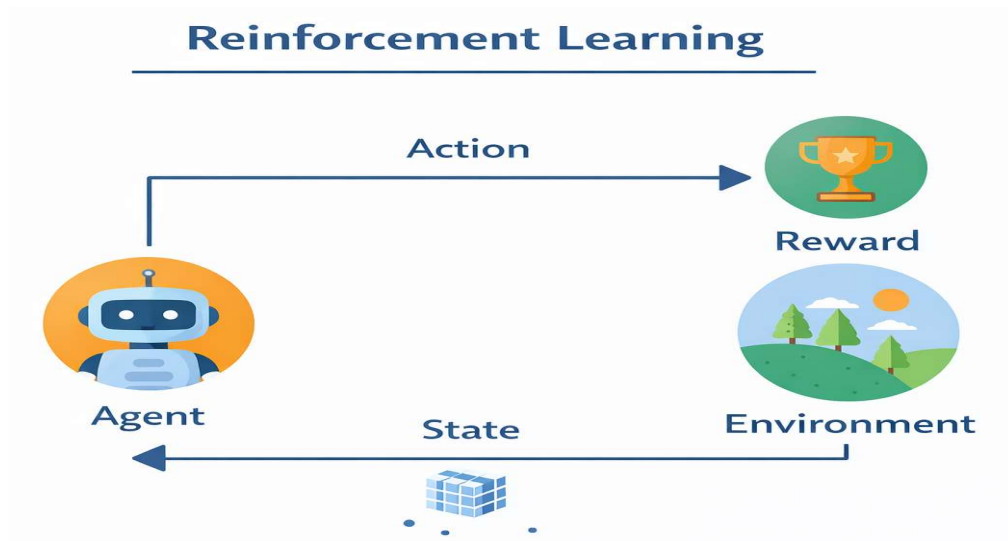


Figure 4.5.2.3 Reinforcement Learning Diagram

4.5.3 Need for Machine Learning in Healthcare

- Large volume of medical data
- Need for early disease detection
- Reduction of human error
- Faster diagnosis
- Improved accuracy

4.5.4 Challenges in Machine Learning

- Data imbalance
- Missing values
- Overfitting
- Feature selection
- Model interpretability

4.6 Algorithms

4.6.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of an outcome using a logistic function. It is simple, efficient, and widely used in medical diagnosis applications.

4.6.2 Random Forest

Random Forest is an ensemble learning algorithm that uses multiple decision trees to make predictions. It improves accuracy and reduces overfitting by averaging the results of multiple trees. Random Forest performs well on large and complex datasets.

4.6.3 Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm that separates data using a hyperplane. It is effective in high-dimensional spaces and is widely used for classification tasks in healthcare applications.

4.6.4 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem. It assumes independence among features and is fast and easy to implement. It works well with large datasets.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Figure 4.6.4.1 Say the Naïve Bayes formula

4.6.5 XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced ensemble machine learning algorithm based on gradient boosting that builds multiple decision trees sequentially to improve prediction accuracy.

In this project, XGBoost is used to classify cervical cancer risk by learning complex patterns from optimized features selected using PSO. It reduces overfitting through regularization and provides high performance, scalability, and fast computation compared to traditional boosting methods.

- An advanced gradient boosting technique.
- Provides high performance and efficiency.

5. SYSTEM DESIGN

5.1 Introduction

System design defines the overall architecture and workflow of the proposed system. It describes how different components interact with each other to achieve the desired output. In this project, system design focuses on structuring the machine learning-based cervical cancer prediction system into various modules for efficient functioning.

5.2 System Architecture

The system architecture illustrates the overall structure of the cervical cancer prediction system. The system takes patient medical and demographic data as input, processes the data using machine learning algorithms, and produces prediction results.

Architecture Description:

- Input Layer: Cervical cancer dataset
- Processing Layer:
 - Data preprocessing
 - Feature selection
 - Model training
- Output Layer: Prediction result (Cancer / No Cancer)

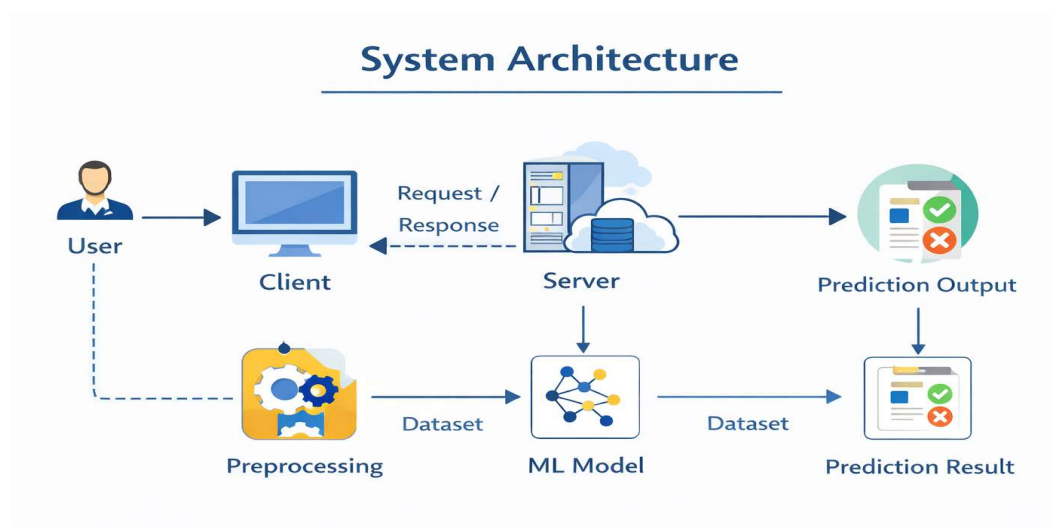


Figure 5.2.1 System Architecture for Cervical Cancer Prediction Using Machine Learning

5.3 System Modules

The proposed system is divided into the following modules:

5.3.1 Data Collection Module

This module is responsible for collecting the cervical cancer dataset. The dataset contains medical and demographic attributes related to cervical cancer risk factors. The data is loaded into the system for further processing.

5.3.2 Data Preprocessing Module

Data preprocessing is a crucial step in machine learning. This module handles missing values, removes irrelevant data, and converts raw data into a suitable format for training the machine learning models. Proper preprocessing improves model accuracy and reliability.

5.3.3 Feature Selection Module

Feature selection identifies the most important attributes that influence cervical cancer prediction. Selecting relevant features reduces dimensionality and improves model performance.

5.3.4 Model Training Module

In this module, machine learning algorithms are trained using the processed dataset. The system learns patterns from the data and builds predictive models.

5.3.5 Prediction Module

The prediction module takes new input data and uses the trained model to predict whether a patient is likely to have cervical cancer or not.

5.4 UML Diagrams

5.4.1 Use Case Diagram

The use case diagram represents the interaction between users and the system.

Actors:

- User
- System

Use Cases:

- Upload Dataset
- Preprocess Data
- Train Model
- Predict Cancer

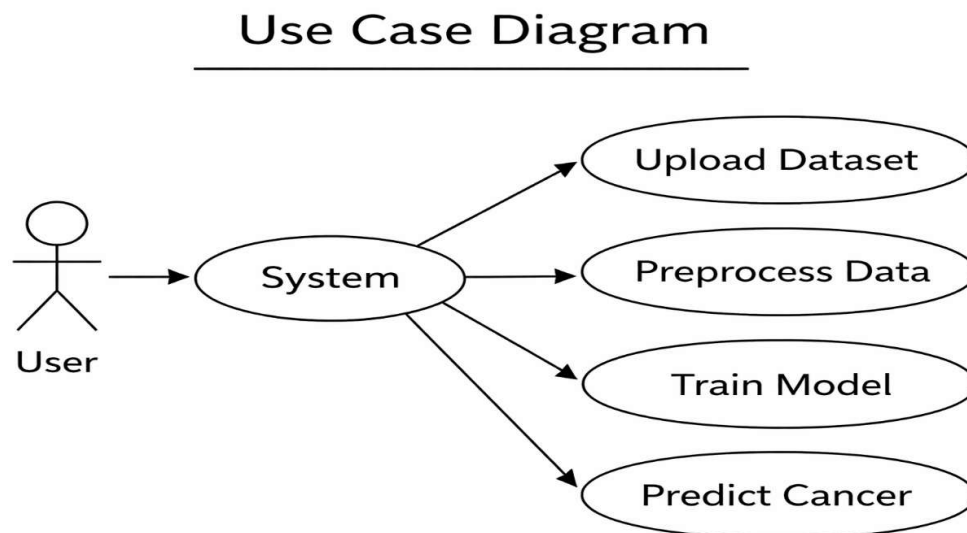


Figure 5.4.1.1 Use Case Diagram for Cancer Prediction System

Figure 5.4.1.1 This image shows a use case diagram representing a cancer prediction system based on machine learning. A stick figure labeled User interacts with the central component labeled System. These include Upload Dataset, Preprocess Data, Train Model, and Predict Cancer. The diagram visually explains the workflow of the application where the user provides data and the system processes it step by step to produce predictions. It highlights how machine learning systems operate starting from data input continuing through data preparation and model training and finally generating cancer.

5.4.2 Activity Diagram

The activity diagram shows the flow of activities in the system.

Steps:

- Start
- Load Dataset
- Preprocess Data
- Train Model
- Predict Result
- End

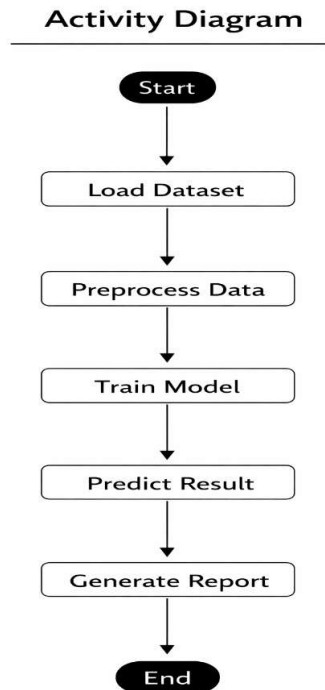


Figure 5.4.2.1 Activity Diagram for Cancer Prediction System

Figure 5.4.2.1 This image illustrates an activity diagram that represents the workflow of a cancer prediction system using machine learning techniques. The process begins with a Start node, followed by sequential steps connected by arrows to show the flow of operations. The first step is Load Dataset, where the system receives the required medical data. Next, the Preprocess Data stage cleans and prepares the dataset for analysis. After that, the Train Model step applies machine learning algorithms to learn patterns from the data.

5.4.3 Sequence Diagram

The sequence diagram represents the interaction between system components over time.

Sequence:

- User → System: Upload data
- System → Model: Train
- Model → System: Prediction
- System → User: Result

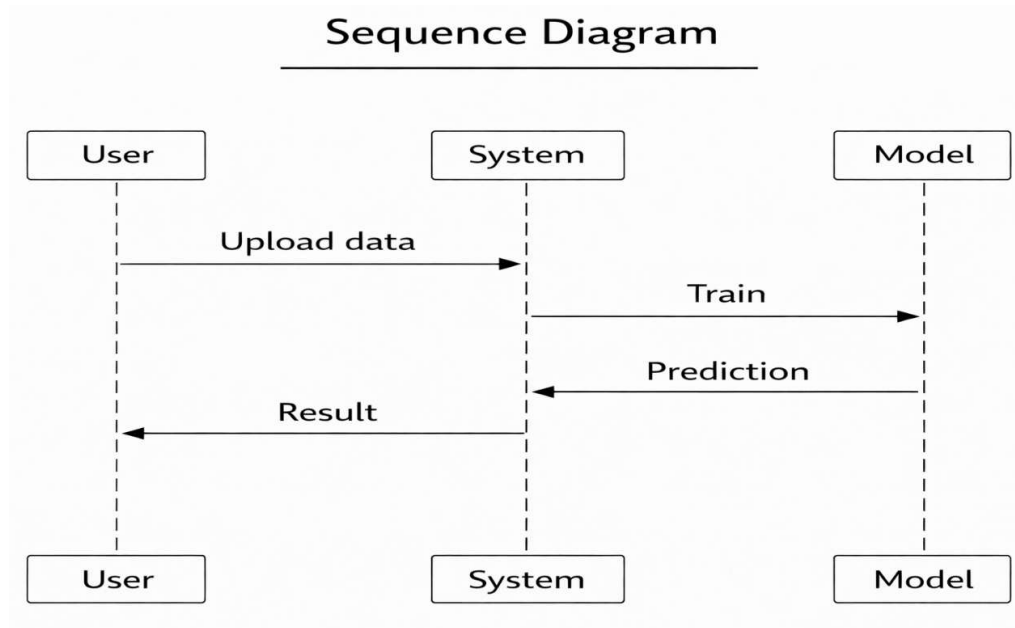


Figure 5.4.3.1 Sequence Diagram for Cancer Prediction System

Figure 5.4.3.1 This image presents a sequence diagram that illustrates the interaction between different components involved in a cancer prediction system. The diagram includes three main entities: User, System, and Model, each represented by vertical lifelines. The process begins when the user uploads data to the system. The system then sends a request to the machine learning model to train using the provided dataset. After training, the model performs prediction and returns the prediction results to the system. Finally, the system sends the generated result back to the user. The diagram clearly shows the step-by-step communication and order of operations among the system components.

5.4.4 Class Diagram

The class diagram shows the structure of classes used in the system.

Classes:

- Dataset
- Preprocessing
- Model
- Prediction

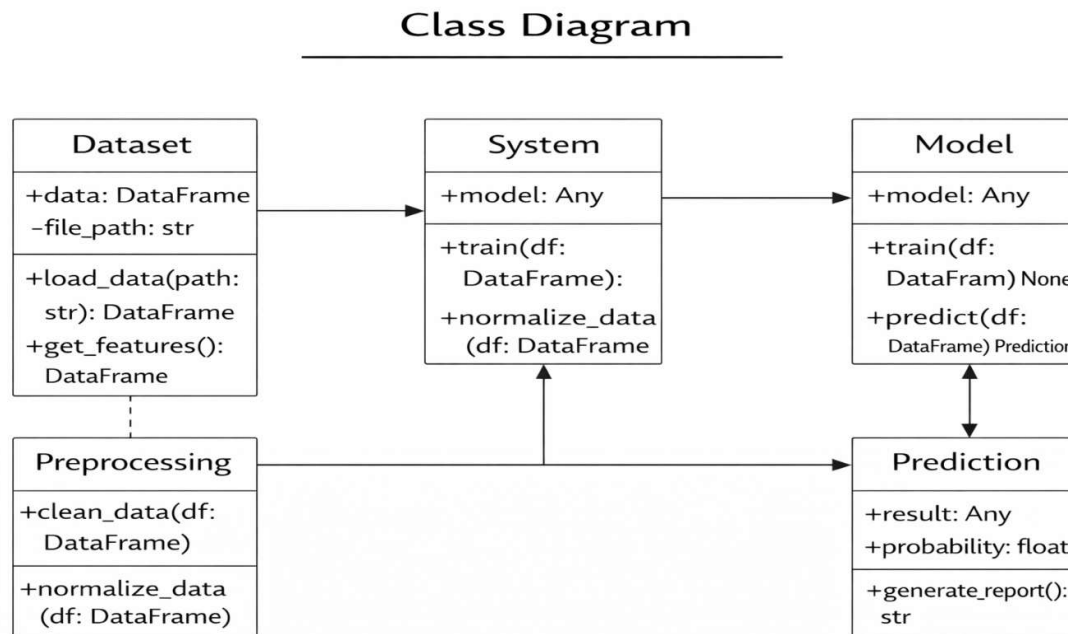


Figure 5.4.4.1 Class Diagram for Cancer Prediction System

Figure 5.4.4.1 This image displays a class diagram representing the structural design of a cancer prediction system. The diagram includes five main classes: Dataset, Preprocessing, System, Model, and Prediction. Each class contains attributes and methods that define its responsibilities. The Dataset class manages data loading and feature extraction, while the Preprocessing class handles data cleaning and normalization. The System class coordinates the overall process and interacts with the machine learning Model class, which performs training and prediction tasks. Finally, the Prediction class stores the prediction result and probability and generates a report. The arrows between classes illustrate relationships and data flow within the system.

5.4.5 Data Flow Diagram (DFD)

The data flow diagram illustrates how data flows through the system.

- Input: Dataset
- Process: Data preprocessing, model training
- Output: Prediction

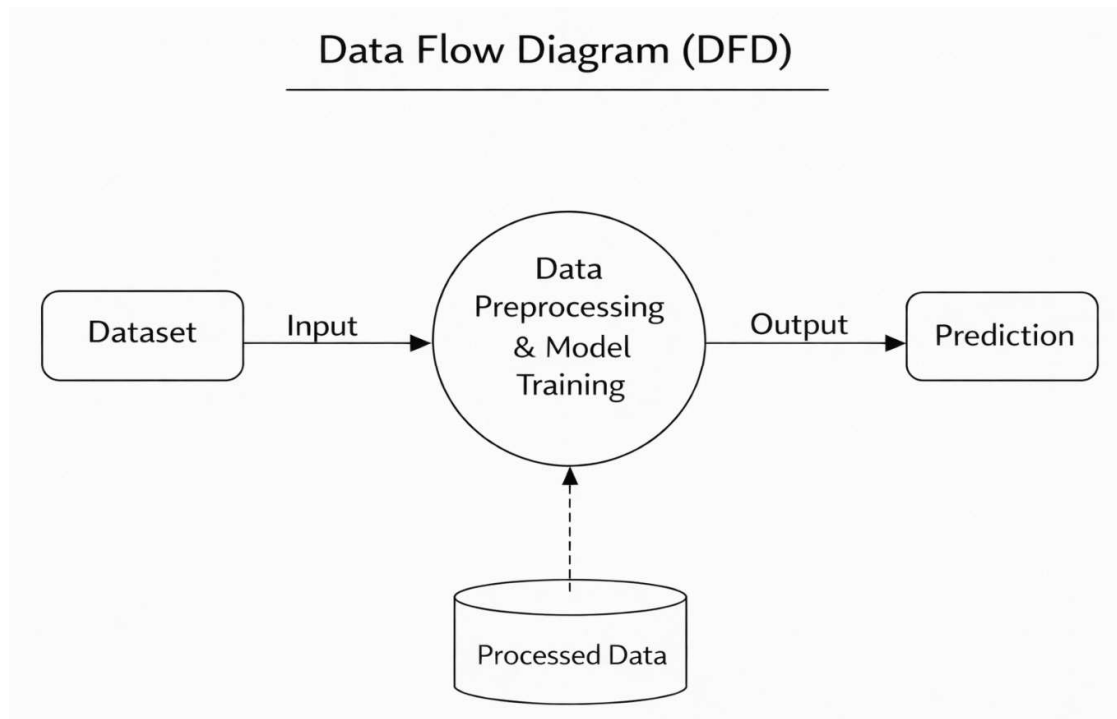


Figure 5.4.5.1 Data Flow Diagram (DFD) for Cancer Prediction System

Figure 5.4.5.1 This image illustrates a Data Flow Diagram (DFD) representing the flow of data in a cancer prediction system. The process begins with the Dataset, which provides the input data to the main processing component labelled Data Preprocessing and Model Training. This central process is responsible for cleaning, preparing, and analyzing the data using machine learning techniques.

6. IMPLEMENTATION

6.1 Introduction

Implementation is the phase where the proposed system is developed and executed using selected tools and technologies. In this project, implementation involves preprocessing the cervical cancer dataset, training machine learning models, and predicting the occurrence of cervical cancer. The implementation is carried out using Python programming language and Google Colab environment.

6.2 Dataset Description

The dataset used in this project is the Cervical Cancer Risk Factors dataset. It contains medical and demographic attributes related to cervical cancer. The dataset includes risk factors such as age, number of pregnancies, sexual behavior, smoking habits, and medical history.

Dataset Characteristics:

- Type: Structured dataset
- Format: CSV
- Number of records: Multiple patient entries
- Target variable: Cervical cancer diagnosis

The dataset is used to train and evaluate machine learning models for prediction.

6.3 Data Preprocessing

Data preprocessing is an essential step to improve the quality of data. The dataset contains missing and inconsistent values, which are handled during preprocessing.

The preprocessing steps include:

- Handling missing values
- Removing irrelevant features
- Encoding categorical variables
- Normalizing numerical data

Proper preprocessing ensures better model accuracy and reliability.

6.4 Feature Selection

Feature selection is performed to identify the most relevant attributes influencing cervical cancer prediction. Selecting important features reduces noise and improves the performance of machine learning models. Feature selection helps in reducing computational complexity and improving accuracy.

6.5 Model Training

After preprocessing and feature selection, machine learning models are trained using the dataset. The dataset is divided into training and testing sets to evaluate the model performance. The models learn patterns from training data and are validated using testing data.

The machine learning algorithms used include:

- Logistic Regression
- Random Forest
- Support Vector Machine
- Naive Bayes
- XGBoost

6.6 Implementation Steps

The following steps describe the implementation process:

1. Import required Python libraries
2. Load the cervical cancer dataset
3. Preprocess the data
4. Perform feature selection
5. Split the dataset into training and testing sets
6. Train machine learning models
7. Evaluate model performance
8. Predict cervical cancer occurrence

6.7 Model Evaluation

The trained models are evaluated using performance metrics such as accuracy, precision, recall, and confusion matrix. These metrics help in understanding the effectiveness of the prediction system.

The evaluation results show that machine learning models provide reliable predictions with good accuracy.

6.8 Code :-

```
from google.colab import drive  
drive.mount('/content/drive')
```

```
!pip install pyswarms imbalanced-learn xgboost
```

```
import pandas as pd  
import numpy as np
```

```
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
from sklearn.ensemble import RandomForestClassifier  
from imblearn.over_sampling import SMOTE
```

```
import pyswarms as ps
```

```

import warnings

warnings.filterwarnings('ignore')


data_path = "/content/drive/MyDrive/Project/Cervical cancel/Cervical_Cancer.csv"

df = pd.read_csv(data_path)

df.head()


# Reload the DataFrame to ensure a clean state for preprocessing

data_path = "/content/drive/MyDrive/Project/Cervical cancel/Cervical_Cancer.csv"

df = pd.read_csv(data_path)


optimizer = ps.single.GlobalBestPSO(

    n_particles=20,

    dimensions=X_train.shape[1],

    options=options

)


cost, best_pos = optimizer.optimize(pso_objective_function, iters=30)


selected_features = best_pos > 0.5


X_train_pso = X_train[:, selected_features]

X_test_pso = X_test[:, selected_features]

```

```
print("Number of Selected Features:", np.sum(selected_features))
```

```
final_model = RandomForestClassifier(  
    n_estimators=300,  
    max_depth=12,  
    random_state=42,  
    n_jobs=-1  
)
```

```
final_model.fit(X_train_pso, y_train)
```

```
y_pred = final_model.predict(X_test_pso)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Final Accuracy:", accuracy * 100)
```

```
print("\nClassification Report:\n")
```

```
print(classification_report(y_test, y_pred))
```

```
print("\nConfusion Matrix:\n")
```

```
print(confusion_matrix(y_test, y_pred))
```

7. TESTING

7.1 Introduction

Testing is an important phase of software development that ensures the system works correctly and meets the specified requirements. In this project, testing is performed to verify the correctness, performance, and reliability of the cervical cancer prediction system. Proper testing helps in identifying errors and improving system quality.

7.2 Types of Testing

7.2.1 Unit Testing

Unit testing involves testing individual modules of the system independently. Each module such as data loading, preprocessing, model training, and prediction is tested separately to ensure correct functionality.

7.2.2 Integration Testing

Integration testing is performed to verify the interaction between different modules. After unit testing, all modules are integrated and tested as a complete system to ensure smooth data flow and correct output generation.

7.2.3 System Testing

System testing evaluates the complete system to ensure that it meets both functional and non-functional requirements. The cervical cancer prediction system is tested using different input datasets to validate prediction accuracy and system stability.

7.3 Performance Testing

Performance testing is conducted to measure the efficiency of the system. The performance of the prediction model is evaluated using metrics such as accuracy, precision, recall, and confusion matrix. These metrics help in determining the effectiveness of the machine learning models used.

7.4 Test Cases

Test Case ID	Description	Expected Result	Status
TC01	Load Dataset	Dataset loaded successfully	Pass
TC02	Data Preprocessing	Data cleaned correctly	Pass
TC03	Model Training	Model trained successfully	Pass
TC04	Prediction	Accurate prediction	Pass

Table 7.4.1 Test Cases for Cancer Prediction System Table

7.5 Evaluation Metrics

The following evaluation metrics are used to measure model performance:

7.5.1 Accuracy

Accuracy measures the percentage of correctly predicted instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Higher accuracy indicates better overall performance.

7.5.2 Precision

Precision measures how many predicted positive cases are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision means fewer false positives.

7.5.3 Recall

Recall measures how many actual positive cases are correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall means fewer false negatives.

7.5.4 F1-Score

F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It balances both precision and recall.

7.5.5 Confusion Matrix

Confusion matrix visually represents classification performance:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 7.2.5.1 Confusion Matrix

8. OUTPUT SCREENS

8.1 Dataset Loading Screen

The dataset is loaded into the system using Python libraries. The dataset preview shows the attributes and records used for training the machine learning models.

	Category	Age	Sex	ALB	ALP	ALT	AST	BIL	CHE
1	0=Blood D	32	m	38.5	52.5	7.7	22.1	7.5	6.93
2	0=Blood D	32	m	38.5	70.3	18	24.7	3.9	11.17
3	0=Blood D	32	m	46.9	74.7	36.2	52.6	6.1	8.84
4	0=Blood D	32	m	43.2	52	30.6	22.6	18.9	7.33
5	0=Blood D	32	m	39.2	74.1	32.6	24.8	9.6	9.15
6	0=Blood D	32	m	41.6	43.3	18.5	19.7	12.3	9.92
7	0=Blood D	32	m	46.3	41.3	17.5	17.8	8.5	7.01
8	0=Blood D	32	m	42.2	41.9	35.8	31.1	16.1	5.82
9	0=Blood D	32	m	50.9	65.5	23.2	21.2	6.9	8.69
10	0=Blood D	32	m	42.4	86.3	20.3	20	35.2	5.46
11	0=Blood D	32	m	44.3	52.3	21.7	22.4	17.2	4.15
12	0=Blood D	33	m	46.4	68.2	10.3	20	5.7	7.36
13	0=Blood D	33	m	36.3	78.6	23.6	22	7	8.56
14	0=Blood D	33	m	39	51.7	15.9	24	6.8	6.46
15	0=Blood D	33	m	38.7	39.8	22.5	23	4.1	4.63
16	0=Blood D	33	m	41.8	65	33.1	38	6.6	8.83
17	0=Blood D	33	m	40.9	73	17.2	22.9	10	6.98
18	0=Blood D	33	m	45.2	88.3	32.4	31.2	10.1	9.78
19	0=Blood D	33	m	36.6	57.1	38.9	40.3	24.9	9.62
20	0=Blood D	33	m	42	63.1	32.6	34.9	11.2	7.01
21	0=Blood D	33	m	44.3	49.8	32.1	21.6	13.1	7.44
22	0=Blood D	33	m	46.7	88.3	23.4	23.9	7.8	9.42
23	0=Blood D	34	m	42.7	65.3	46.7	30.3	23.4	10.95
24	0=Blood D	34	m	43.4	46.1	97.8	46.2	11.3	7.99
25	0=Blood D	34	m	40.5	32.4	29.6	27.1	5.8	10.5
26	0=Blood D	34	m	44.8	77.7	36.9	31	19.5	10.51
27	0=Blood D	34	m	42.6	27	21.4	21.7	7.2	8.15
28	0=Blood D	34	m	29	41.6	29.1	16.1	4.8	6.82
29	0=Blood D	34	m	44.6	84.1	19.6	29.8	5.8	7.6
30	0=Blood D	34	m	46.8	61.7	24.5	24.2	23.1	10.99
31	0=Blood D	34	m	41.8	75.8	30.9	35.5	6.1	9.97
32	0=Blood D	34	m	46.1	70.6	35.8	30	7.6	7.7
33	0=Blood D	34	m	43.6	58.9	47.1	31.1	18.5	9.14
34	0=Blood D	35	m	37.5	69.8	37.1	25	7.8	11.66
35	0=Blood D	35	m	42.1	68.3	37.2	56.2	11.1	9.3
36	0=Blood D	35	m	44.7	79.3	53.5	30.8	9.7	11.39
37	0=Blood D	35	m	41.5	115.1	24.1	30.4	5.7	9.41
38	0=Blood D	35	m	48.7	72.7	24.1	31	45.1	9.4
39	0=Blood D	35	m	47.3	92.2	30.7	25.7	6.6	11.58
40	0=Blood D	35	m	44.5	70.3	26.2	25.1	5.1	10.12
41	0=Blood D	35	m	47.4	54.5	18.6	21.6	10.3	8.1
42	0=Blood D	35	m	51	82.7	29.3	26.8	8.7	12.32
43	0=Blood D	35	m	27.8	99	30.7	27.8	9.4	6.8
44	0=Blood D	36	m	46.1	58.5	26.8	25.3	6	6.61
45	0=Blood D	36	m	45.5	57.6	22.5	19.5	7.5	5.28
46	0=Blood D	36	m	41.7	77.2	103.6	46.9	10.4	12.21

Figure: 8.1.1 Datasets

Figure 8.1.1 This image displays a tabular dataset related to medical or health records, likely used for disease prediction or analysis. The table contains several columns including Category, Age, Sex, ALB, ALP, ALT, AST, BIL, and CHE, which represent patient information and biochemical test results. Each row corresponds to an individual patient record with values for these medical parameters. The Category column appears to indicate the health status or type of case, while Age and Sex describe demographic details. The remaining columns represent laboratory measurements of liver-related enzymes and proteins. Such datasets are commonly used in machine learning models for medical diagnosis and disease prediction.

8.2 Data Preprocessing Output

This screen shows the preprocessing steps such as handling missing values, cleaning data, and preparing the dataset for training. Proper preprocessing improves the performance of machine learning models.

```
In [2]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
        dataset = pd.read_csv("cancer_data.csv")
        X = dataset.drop('id', 'diagnosis', axis=1)
        y = dataset[_diagnosis'].map({'M': 1, 'B': 0})
        X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
        random_state=42)
        scaler = StandardScaler()
        X_train_scaled = scaler.fit_transform(X_train)
        X_test_scaled = scaler.transform(X_test)

Out[3]: X_train_scaled[:5]

In [3]: X_train_scaled[:5]

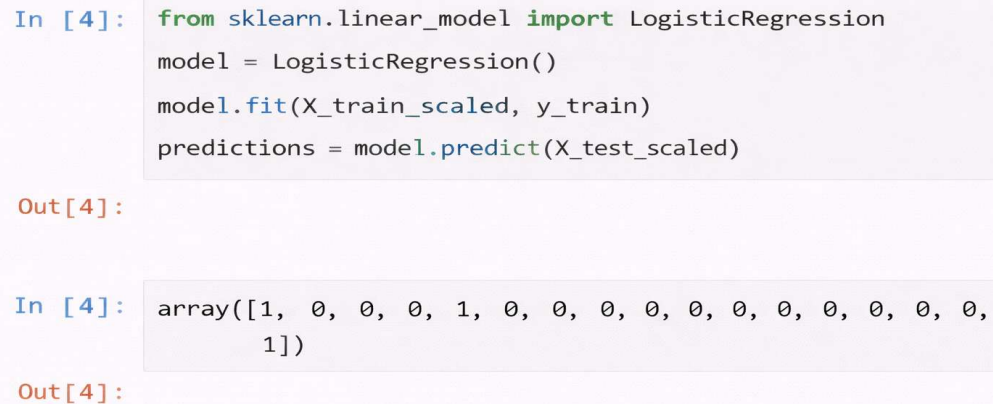
Out[3]: -3.8689524 -6.5986992 -5.6245842  6.8669768  4.913395 -4.95764675  :]
        -4.0776098 -6.1433520  3.5988246 -4.6326994  7.707000 -2.1635270  :]
        -5.2812954 -5.3368399  3.9176231 -1.6886905  6.199645 -4.70552012  :]
        -0.3629229 -1.1234098 - 0.0388906 -6.3882173 - 5.104950  1.68238537  :]
        -4.3096675 -C.6915381- 0.2636523  0.1632073  5.548650 -6.0879291  :]
        [-3.6889520 -3.5368992 -5.8245842 - 8.8659168 -4.912395 -4.95764673  .]
```

Figure 8.2.1 Datasets

Figure 8.2.1 This image shows a Jupyter Notebook screenshot demonstrating the data preprocessing stage in a machine learning workflow. The code uses Python libraries such as pandas and scikit-learn to prepare a cancer dataset for model training. First, the dataset is loaded using `pd.read_csv()`. Then unnecessary columns like `id` and `diagnosis` are removed from the feature set. The target variable is converted into numerical form using mapping. The data is split into training and testing sets using `train_test_split()`. After that, `StandardScaler` is applied to normalize the feature values. The output section displays a preview of the scaled training data, represented as standardized numerical values.

8.3 Model Training Output

This screen displays the output of the machine learning model training process. It includes training logs and confirmation of successful model creation.



```
In [4]: from sklearn.linear_model import LogisticRegression
        model = LogisticRegression()
        model.fit(X_train_scaled, y_train)
        predictions = model.predict(X_test_scaled)

Out[4]:
```



```
In [4]: array([1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        1])

Out[4]:
```

Figure 8.3.1 Model Training Output

Figure 8.3.1 This image displays a Jupyter Notebook screenshot illustrating the model training and prediction step in a machine learning process. The Python code imports `LogisticRegression` from the `scikit-learn` library to build a classification model. A logistic regression model is created and trained using the scaled training dataset (`X_train_scaled`) along with the corresponding training labels (`y_train`). After training the model with the `fit()` function, predictions are generated on the test dataset (`X_test_scaled`) using the `predict()` method. The output section shows an array of predicted values, where numbers such as 1 and 0 represent different classification outcomes for the test samples in the dataset.

8.4 Prediction Result Screen

This screen displays the final prediction result indicating whether the patient is likely to have cervical cancer or not. The prediction output is generated using the trained machine learning model.

The screenshot displays a web-based interface titled "Cervical Cancer Prediction System". Below the title is a subtitle: "This system uses machine learning to predict the risk of cervical cancer based on blood test results and patient information." The interface features a form with the following fields:

- Age:
- Sex:
- Albumin (ALB) (g/L):
- Alkaline Phosphatase (ALP) (U/L):
- Alanine Aminotransferase (ALT) (U/L):
- Aspartate Aminotransferase (AST) (U/L):
- Bilirubin (BIL) (μmol/L):
- Cholinesterase (CHE) (kU/L):
- Cholesterol (CHOL) (mmol/L):
- Creatinine (CREA) (μmol/L):
- Gamma-Glutamyl Transferase (GGT) (U/L):
- Protein (PROT) (g/L):

Below the form are two buttons: "Predict Risk" (blue) and "Load Random Sample" (purple). At the bottom, there is a section labeled "Prediction Result:" with a light blue background and a vertical line on the left side.

Fig: 8.4.1 Empty Output Screen

Figure 8.4.1 : The image shows a web-based interface titled “Cervical Cancer Prediction System.” The system is designed to predict the risk of cervical cancer using machine learning techniques based on patient information and laboratory test values. The interface has a clean and modern layout with a light-colored background and a centered form container.

At the top, a heading clearly displays the system name, followed by a short description explaining that the model uses blood test results and patient data to assess cervical cancer risk.

Below the heading, a structured input form is presented. The form includes multiple labeled fields such as Age and Sex (with a dropdown selection option). It also contains various

biochemical parameters including Albumin (ALB), Alkaline Phosphatase (ALP), Alanine Aminotransferase (ALT), Aspartate Aminotransferase (AST), Bilirubin (BIL), Cholinesterase (CHE), Cholesterol (CHOL), Creatinine (CREA), Gamma-Glutamyl Transferase (GGT), and Protein (PROT). Each parameter has an input box where users can enter numerical values.

At the bottom of the form, two buttons are provided: “Predict Risk” and “Load Random Sample.” Beneath these buttons, a “Prediction Result” section is displayed, where the system outputs the final classification result after processing the input data.

The screenshot displays a web-based application titled "Cervical Cancer Prediction System". Below the title is a subtitle: "This system uses machine learning to predict the risk of cervical cancer based on blood test results and patient information." The form contains the following fields and values:

- Age: 25
- Sex: Male (dropdown menu)
- Albumin (ALB) (g/L): 42
- Alkaline Phosphatase (ALP) (U/L): 38.2
- Alanine Aminotransferase (ALT) (U/L): 63.3
- Aspartate Aminotransferase (AST) (U/L): 187.7
- Bilirubin (BIL) (μmol/L): 14
- Cholinesterase (CHE) (kU/L): 6
- Cholesterol (CHOL) (mmol/L): 4.28
- Creatinine (CREA) (μmol/L): 66.9
- Gamma-Glutamyl Transferase (GGT) (U/L): 40.2
- Protein (PROT) (g/L): 70.5

At the bottom of the form are two buttons: "Predict Risk" (blue) and "Load Random Sample" (purple). Below these buttons is a "Prediction Result" section with the following content:

Prediction Result:
Risk Level: High Risk (Hepatitis Category)
Risk Score: 69/100

Interpretation:
Your values indicate possible hepatitis or liver dysfunction.
Consult with a healthcare professional for proper diagnosis.

Figure 8.4.2 Predicted Output Screen

Fig: 8.4.2 : The image displays a web-based application interface titled “Cervical Cancer Prediction System.” The interface contains a structured form designed to collect patient details and laboratory test values for disease risk prediction using machine learning. The form includes fields such as Age (25), Sex (Male), Albumin (42 g/L), Alkaline Phosphatase (38.2 U/L), Alanine Aminotransferase (63.3 U/L), Aspartate Aminotransferase (187.7 U/L), Bilirubin (14 μmol/L),

Cholinesterase (6 kU/L), Cholesterol (4.28 mmol/L), Creatinine (66.9 μ mol/L), Gamma-Glutamyl Transferase (40.2 U/L), and Protein (70.5 g/L).

Below the input fields, two buttons are visible: “Predict Risk” and “Load Random Sample.” The prediction output section displays the result: “Risk Level: High Risk (Hepatitis Category)” with a Risk Score of 69/100. An interpretation message explains that the values indicate possible hepatitis or liver dysfunction and advises consulting a healthcare professional for proper diagnosis. The layout is clean and user-friendly, with a centered card-style design and clear section separation between input fields and prediction results.

9. CONCLUSION

This project titled “Cervical Cancer Prediction Using Machine Learning” successfully demonstrates the application of machine learning techniques in the healthcare domain. The proposed system analyzes medical and demographic data to predict the occurrence of cervical cancer at an early stage.

The system uses machine learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine, and Naive Bayes to classify patient data. Data preprocessing and feature selection techniques were applied to improve model accuracy and reliability. The experimental results show that machine learning-based prediction systems can assist medical professionals by providing fast and accurate results.

The proposed system is cost-effective, efficient, and reduces dependency on manual diagnosis. It acts as a decision support tool and helps in early detection, which can significantly improve patient survival rates.

10. FUTURE SCOPE

Although the proposed system performs effectively, there is scope for further improvement. The future enhancements of this project include:

- Integration of deep learning models for higher accuracy
- Real-time data collection from hospitals
- Deployment as a web-based or mobile application
- Integration with electronic health record systems
- Use of advanced feature selection techniques

These enhancements can further improve prediction accuracy and make the system more useful in real-world healthcare environments.

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RESEARCH ARTICLE

Machine Learning-Assisted Cervical Cancer Prediction Using Particle Swarm Optimization for Improved Feature Selection and Prediction

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This work was supported in part by the South African National Research Foundation under Grant 141951, Grants nos. 137951 and AJCR230704126719120106.

ABSTRACT Cervical cancer is a common and deadly disease that affects women worldwide. Early diagnosis and treatment can improve the survival and quality of life of patients. Machine learning techniques can help to analyze complex and high-dimensional data related to cervical cancer and provide accurate and reliable predictions. However, selecting the most relevant and informative features from the data is a challenging task that affects the performance and interpretability of machine learning models. This paper proposes a novel method that uses particle swarm optimization (PSO) to perform feature selection and optimization for cervical cancer prediction. PSO is a bio-inspired algorithm that mimics the social behavior of a swarm of particles that search for the optimal solution in the feature space. The use of PSO to select the best subset of features that maximize the classification accuracy of eight machine learning models: Support Vector Machines (SVM), Gaussian Naive Bayes (GNB), Random Forests (RF), Decision Trees (DT), Extreme Gradient Boosting (XGB), Linear Regression (LR), Adaptive Boosting (AdaBoost), and K-nearest neighbor (KNN). To evaluate the method, a publicly available dataset was used, the Cervical Cancer Risk Factors Dataset (CCRFD). Then, compare the results with several state-of-the-art methods that use different feature selection techniques and ML algorithms. The experimental results show that the method achieves superior performance in terms of feature reduction rate, accuracy, precision, and AUC. Specifically, the Adaboost-PSO model performed best in terms of feature reduction rate with a reduction of rate 100% while the RF-PSO model performed best in terms of accuracy and precision, with an accuracy of 98% and precision of 100%.

INDEX TERMS Machine learning, feature selection, cervical cancer, particle swarm optimization.

I. INTRODUCTION

Cervical cancer is a type of cancer that affects the cervix, which is the lower part of the uterus that connects to the vagina. Cervical cancer is one of the most common and deadly cancers among women worldwide, especially in developing countries. According to the World Health Organization (WHO), cervical cancer is the fourth most frequent cancer in women, with an estimated 604,000 new cases and 342,000 deaths in 2020 [1]. Global cancer statistics [2] show that cervical cancer affects 493,000 new patients every

year, making up 15% of all female cancers. This disease is prevalent in developing countries [3], where it has a high mortality rate of 83%. It is especially common in African countries [4], [5]. Cervical cancer is caused by persistent infection with certain types of human papillomavirus (HPV), which can be transmitted through sexual contact. Other risk factors for cervical cancer include smoking, multiple sexual partners, early sexual activity, long-term use of oral contraceptives, and weakened immune system. Some of the symptoms of cervical cancer include abnormal vaginal bleeding, pain during sex, pelvic pain, and vaginal discharge. Cervical cancer can be diagnosed by various methods, such as Pap smear test, HPV test, colposcopy, biopsy, etc. Cervical

The associate editor coordinating the review of this manuscript and approving it for publication was Essam A. Rashed^{ID}.

cancer can be treated by various methods, such as surgery, radiation therapy, chemotherapy, immunotherapy, etc [6], [7], [8], [9].

Machine learning has been widely applied to various domains, such as computer vision, natural language processing, speech recognition, etc. Machine learning can also be used for analyzing complex and high dimensional biomedical data and making accurate predictions for diagnosis, prognosis, and survival of various diseases, such as cancer. Machine learning can help to improve the quality and efficiency of healthcare services and reduce the cost and time of medical decision making [10], [11], [12], [13]

Particle swarm optimization (PSO) is an evolutionary computation technique that simulates the social behavior of bird flocks or fish schools. PSO consists of a set of particles that represent potential solutions to an optimization problem. Each particle has a position vector and a velocity vector in a multidimensional search space. Each particle also has a personal best position and a global best position based on their fitness values. The particles update their velocities and positions according to their own and their neighbors' best positions until they converge to the optimal solution. PSO can be used for feature selection, which is a process of selecting a subset of relevant features from a large feature space. Feature selection can improve the performance and efficiency of machine learning models by reducing the computational cost, enhancing the generalization ability, and avoiding the curse of dimensionality. The main objective of this article is to propose a machine learning assisted cervical cancer prediction model using particle swarm optimization for feature selection. The main contributions of this article are as follows:

- 1) Particle Swarm optimization is used to select a subset of relevant features from the original feature space, which includes 15 features related to demographic information, habits, sexual behavior, gynecological history, and HPV infection status.
- 2) Eight machine learning models, Logistic Regression (LR), Support Vector Classifier (SVC), Random Forest (RF), Gaussian Naive Bayes (GNB), Decision Trees (DT), Extreme Gradient Boosting (XGB), K-Nearest Neighbors (KNN), and Adaptive Boosting (AdaBoost) are combined with PSO-selected features to enhance the performance of cervical cancer prediction.
- 3) The results indicate that "Age," "First sexual intercourse," and "Number of Pregnancies" are the top-ranked features for cervical cancer prediction.

A. RELATED WORK

Machine learning has been widely applied to various domains, such as computer vision, natural language processing, speech recognition, etc. Machine learning has also been used for analyzing complex and high-dimensional biomedical data and making accurate predictions for diagnosis, prognosis, and survival of various diseases, such as cancer. Machine learning can help to improve the quality and efficiency of healthcare services and reduce the cost and time of

medical decision making. Many studies in the literature have used machine learning for cervical cancer detection, diagnosis, prognosis, and survival prediction. Mehmood et al. [14] proposed a machine learning-assisted cervical cancer detection model using particle swarm optimization for feature selection. They used a real-world dataset of cervical cancer risk factors obtained from UCI Machine Learning Repository to train and test their model. They used particle swarm optimization to select a subset of relevant features from the original feature space, which consists of 36 features related to demographic information, habits, sexual behavior, gynecological history, HPV infection status, etc. They used four machine learning models: logistic regression (LR), support vector machines (SVM), random forest (RF), and artificial neural network (ANN) to predict whether a patient has cervical cancer or not based on the selected features. They evaluated the performance of their model using various metrics, such as accuracy, precision, recall, F1-score, etc., and compared it with other existing models in the literature. They found that their model outperformed other models and achieved an accuracy of 93.6%, mean squared error (MSE) error of 0.07111, false-positive rate (FPR) of 6.4% and false-negative rate (FNR) of 100%, [14].

Alsallat et al. [15] used machine learning to analyze cervical cancer data obtained from Kaggle. They used three machine learning models: decision tree (DT), k-nearest neighbor (KNN), and naive Bayes (NB) to classify the patients into two groups: healthy or with cervical cancer. They used 10-fold cross-validation to evaluate the performance of their models using accuracy as the metric. They found that DT achieved the highest accuracy of 97%, followed by KNN with 95%, and NB with 93%, [15].

Aljrees [16] addresses the challenge of missing data in cervical cancer detection by proposing a stacked ensemble model combining Extreme Gradient Boosting (XGB), Random Forest (RF), and Extra Tree Classifier (ETC). The study uses a K-Nearest Neighbors (KNN) Imputer to manage missing values, enhancing model reliability. The approach is evaluated using a cervical cancer dataset from Kaggle, exploring three scenarios: deletion of missing values, KNN imputation, and Principal Component Analysis (PCA) imputation. The ensemble model achieved an accuracy of 99.41%, with precision at 0.98%, recall at 0.96%, and an F1 score of 0.97%, outperforming traditional methods. This research contributes significantly to medical diagnostics by improving early cervical cancer detection, offering a robust solution for clinical application.

Uddin et al. [17] present an ensemble machine learning approach to enhance cervical cancer prediction. The study focuses on early detection and risk assessment using the UCI Machine Learning Repository's "Risk Factors" dataset, which includes 858 samples and 36 attributes. The authors employed multiple classifiers, including SVM, RF, and KNN, and used a hybrid feature selection process incorporating PCA, XGBoost, and SelectKBest to identify key predictive features. The ensemble model, combining Random Forest

and Multilayer Perceptron with Random Oversampling, achieved a notable accuracy of 99.19%, highlighting its effectiveness in managing high-dimensional and imbalanced data. This approach shows promise for broader applications in medical diagnostics.

Edafetanure-Ibeh [18] implemented machine learning methods for cervical cancer prediction using a publicly available dataset from the UCI Machine Learning Repository. The dataset, collected at the Hospital Universitario de Caracas in Venezuela, includes medical history, risk factors, demographics, and test findings for cervical cancer diagnosis. The study evaluated techniques such as Random Forest, Naive Bayes, SVM (linear kernel), KNN, Logistic Regression, and XGBoost. XGBoost proved to be the most effective model with a 95.4% accuracy, followed by Random Forest at 98%. SVM and Logistic Regression achieved 95% and 94% accuracy, respectively, while KNN reached 97%. Gaussian Naive Bayes had the lowest accuracy at 85%.

Ding et al. [19] used machine learning to predict cervical cancer survival using a dataset obtained from SEER. They used four machine learning models: Cox proportional hazards (CPH), random survival forest (RSF), support vector regression (SVR), and deep neural network (DNN) to predict the survival time and risk score of the patients based on 16 features related to demographic information, tumor characteristics, treatment modalities, etc. They used the concordance index (C-index) and integrated Brier score (IBS) as the evaluation metrics. They found that DNN achieved the highest C-index of 0.72 and the lowest IBS of 0.11, [19]. These studies have shown promising results on using machine learning for cervical cancer prediction, but they also have some limitations, such as:

- 1) They may not consider all the relevant features or use appropriate feature selection methods to reduce the dimensionality and complexity of the feature space.
- 2) They may not use suitable machine learning models or optimize their hyperparameters to achieve better performance and generalization.
- 3) They may not use adequate evaluation metrics or validation methods to assess the reliability and robustness of their models.
- 4) They may not compare their results with another state of the art studies or provide sufficient explanations and interpretations of their results.

Therefore, there is a need for a comprehensive and systematic study on machine learning-assisted cervical cancer prediction using particle swarm optimization for feature selection. This study aims to fill this gap by proposing a novel and efficient model that can select the most relevant features and predict the cervical cancer status of the patients with high accuracy and reliability

II. MACHINE LEARNING MODELS

We use eight machine learning models to predict whether a patient has cervical cancer or not based on the selected

features. The machine learning models are briefly introduced in the following subsections.

A. LOGISTIC REGRESSION

LR is a linear model that uses a logistic function to model the probability of a binary outcome. LR estimates the coefficients of the features using maximum likelihood estimation and predicts the outcome using a threshold value [20].

B. SUPPORT VECTOR MACHINES

SVM is a kernel-based model that uses a hyperplane to separate the data into two classes. SVM finds the optimal hyperplane that maximizes the margin between the classes and minimizes the classification error. SVM can also use different kernel functions to map the data into higher-dimensional spaces and create nonlinear decision boundaries. In this work, we use Support Vector Classifier (SVC) for our classification tasks [21], [22], [23].

C. RANDOM FOREST

RF is an ensemble model that uses multiple decision trees to make predictions. RF builds each decision tree using a random subset of features and a random subset of instances. RF then aggregates the predictions of all the trees using majority voting or averaging [24], [25], [26].

D. K-NEAREST NEIGHBOURS

K-nearest neighbour, or KNN, is a supervised learning algorithm that can be used for both classification and regression problems. It is based on the idea that similar data points tend to have similar labels or values. KNN does not require any explicit training, but rather stores the entire training data and uses it to make predictions for new data points. It defines a distance metric to measure the similarity between data points, chooses a value for k , which is the number of nearest neighbours to consider, and finds the k closest training data points for each new data point. Then, it assigns the label that is most frequent among the k nearest neighbours for classification, or the average of the values of the k nearest neighbours for regression, as the prediction for the new data point. It is simple and intuitive to understand and implement, it can handle nonlinear and complex data patterns and can be easily adapted to different scenarios. However, it can be computationally expensive and slow, sensitive to noise, outliers, and irrelevant features, and suffer from the curse of dimensionality [27], [28], [29], [30].

E. DECISION TREES

A decision tree classifier is a supervised machine learning algorithm that assigns class labels to data instances based on a set of rules that are derived from the attributes of the data. The algorithm constructs a hierarchical structure that resembles a tree, where each internal node represents a test or a condition on an attribute, each branch represents an outcome or a decision, and each terminal node represents a

class label or a prediction. The algorithm learns the rules by recursively partitioning the data into smaller subsets based on the values of the attributes until a stopping criterion is met, such as the maximum depth of the tree or the minimum number of instances in a node [31], [32], [33].

F. ADAPTIVE BOOSTING

Adaboost is a popular and powerful ensemble learning technique that combines multiple weak classifiers to form a strong classifier. It adapts the weights of the weak classifiers and the training samples at each iteration to improve the performance of the final classifier [34]. It is simple and easy to implement, fast and scalable and can perform well in some domains where the features are independent and normally distributed. It can also effectively combine strong base learners, producing an even more accurate model [34], [35]. However, it may perform poorly on domains where the features are correlated or have non-normal distributions; it may suffer from overfitting if the noise level is high or the number of iterations is too large and may encounter the zero frequency problem, where it assigns zero probability to unseen feature values [34], [35], [36].

G. EXTREME GRADIENT BOOST

Gradient tree boosting is a machine-learning technique that can perform well on various real-world problems. It is a method of combining multiple weak learners, usually decision trees, into a strong learner by iteratively adding new trees that minimize a loss function. Gradient tree boosting has been shown to achieve state of the art results on many standard classification benchmarks. XGBoost is a scalable and efficient implementation of gradient tree boosting. It is an open-source software library that provides a fast and flexible version of the gradient boosting algorithm. XGBoost follows the same principle as gradient boosting, but it only uses decision trees as its base learners. Therefore, it uses a different loss function that can control the complexity of the trees, such as the number of leaves, the depth, or the weight. XGBoost also uses a regularized model to prevent overfitting and improve generalization [37], [38], [39], [40].

H. GAUSSIAN NAIVE BAYES

A Gaussian Naive Bayes classifier is a type of supervised machine learning algorithm that applies Bayes' theorem to classify data based on the assumption that the features are independent and normally distributed. It calculates the prior probabilities of each class, the mean and standard deviation of each feature for each class, and the likelihood of each feature given each class. Then, it predicts the class with the highest posterior probability as the output. It is simple and easy to implement, fast and scalable and can perform well on some domains where the features are indeed independent and normally distributed. However, it may perform poorly on domains where the features are correlated or have

non-normal distributions and may suffer from zero frequency problems [41], [42], [43], [44].

III. RESEARCH METHODOLOGY

This section describes the dataset, the preprocessing steps, the particle swarm optimization algorithm, the machine learning models, and the evaluation metrics used in our study. Figure 1 shows the flow chart of our proposed PSO-based feature selection for cervical cancer prediction.

A. DATASET

The dataset used in the study is obtained from UCI Machine Learning Repository 1, which is a collection of datasets for machine learning research. The dataset contains 858 instances of patients who underwent cervical cancer screening at Hospital Universitario de Caracas in Venezuela. The dataset consists of 36 features related to demographic information, habits, sexual behavior, gynecological history, HPV infection status, etc. The features are a mixture of nominal, ordinal, and numerical variables. The target variable is Biopsy, which indicates whether the patient has cervical cancer (1) or not (0). The dataset is imbalanced, as only 55 instances 6.4% belong to the positive class and 803 instances 93.6% belong to the negative class.

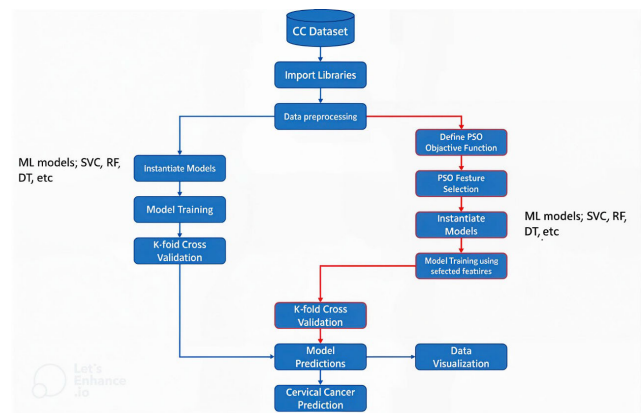


FIGURE 1. Flow chart of cervical cancer prediction using PSO for improved feature selection.

B. IMPORT LIBRARIES

In building the ML models for cervical cancer prediction, some Python libraries were used. These libraries include libraries for preprocessing the data, defining the PSO parameters, instantiating the ML models, training the models, and so on.

C. PREPROCESSING

Medical data have multiple features, each with various kinds of values. The data quality may be affected by noise, outliers, missing or duplicate data, and unrepresentative or biased data [45]. To improve the raw data for further analysis, preprocessing steps are needed to focus on the data

preparation. The preprocessing steps applied to the dataset are as follows:

- 1) **cleaning:** Instances with missing values or outliers are removed from the dataset to ensure the quality and accuracy of the analysis.
- 2) **Data normalization:** We normalize the numerical features to have zero mean and unit variance, as this can improve the performance and convergence of the machine learning models.
- 3) **Data imputation:** We impute the missing values of the nominal and ordinal features using the most frequent value of each feature, as this can preserve the distribution and mode of the data.
- 4) **Data balancing:** This is required as it can prevent the machine learning models from being biased towards the majority class and improve their generalization ability.

D. INSTANTIATE MODELS

The ML models for prediction were instantiated by calling the appropriate library and passing the parameters for each ML model.

E. DEFINE PSO OBJECTIVE FUNCTION

Using the PSO algorithm requires defining the objective function that controls the boundaries and convergence characteristics of the PSO algorithm. Also, the parameters of the PSO feature selection are also passed to the defined PSO objective function.

F. PSO FEATURE SELECTION

The defined PSO objective function and parameters are used to perform feature selection on the preprocessed CCV dataset.

G. MODEL TRAINING

After the completion of the PSO feature selection process, a new dataset is created based on the selected feature and the ML models instantiated are trained using the PSO-selected features. Each model is trained using their tailored specific parameters and hyper-parameters.

H. K-FOLD CROSS VALIDATION

K-fold cross-validation is a technique for K-fold Cross Validation evaluating the performance of a machine learning model on a given dataset. It involves splitting the dataset into k subsets, or folds, of roughly equal size. Then, for each fold, the model is trained on the remaining k-1 folds and tested on the current fold. This process is repeated k times so that each fold is used as a test set once. The average of the k test results is then used as an estimate of the model's accuracy [46], [47], [48], [49]. This technique has several advantages over other methods of model evaluation, such as holdout validation or leave one out validation. Some of these advantages are [50], [51]:

- 1) It reduces the variance of the test results since the model is tested on different subsets of the data.
- 2) It makes use of all the available data for both training and testing, which is especially useful when the dataset is small or scarce.
- 3) It allows for tuning hyperparameters, such as the number of epochs, learning rate, or regularization, by comparing the test results for different values of these parameters.

K-fold cross-validation is applied with k set at 10 to enhance model performance across datasets.

I. MODEL PREDICTIONS

The trained models are used to make predictions of cervical cancer presence or not and these are used for insights in diagnosis and treatment for the patient(s).

J. DATA VISUALIZATION

In order to visualize the performance of the models using the performance metrics, the training results are visually presented. Specifically, the Confusion Matrix was plotted that show the number of True Positives, True Negatives, False Positives and False Negatives. Also, the visual plots of the metrics were presented for better visual analysis.

K. FEATURE SELECTION

Feature selection is a process of choosing a subset of relevant features from a large data set for building predictive models. It can improve the performance, interpretability, and generalization of the models, as well as reduce the computational cost and complexity. There are different types of feature selection methods, such as filter, wrapper, and embedded methods. Filter methods use statistical tests to rank the features based on their correlation. With the target variable. Wrapper methods use a search algorithm to find the optimal subset of features that maximizes the accuracy of a given model. Embedded methods use regularization techniques to penalize or shrink the coefficients of irrelevant features [52], [53].

L. PARTICLE SWARM OPTIMIZATION

Particle swarm optimization (PSO) is an evolutionary computation technique that simulates the social behavior of bird flocks or fish schools and has been used extensively in the literature [54], [55], [56], [57], [58], [59]. PSO consists of a set of particles that represent potential solutions to an optimization problem. Each particle has a position vector x_i and a velocity vector v_i in a D-dimensional search space, where i is the index of the particle, and D is the number of features. Each particle also has a personal best position p_i , which is the best position that the particle has visited so far, and a global best position g , which is the best position among all the personal best positions in the swarm. The Particle Swarm Optimization (PSO) algorithm was chosen due to its computational efficiency and simplicity in handling

feature selection. PSO is particularly suited for problems involving a large search space, like feature selection, because it effectively balances exploration and exploitation of the solution space. Unlike other optimization techniques, PSO has fewer parameters to adjust, is easy to implement, and has been shown to converge faster on feature selection tasks. Additionally, its ability to avoid local optima makes it suitable for our high-dimensional feature space. The fitness function $f(x_i)$ evaluates the quality of each position x_i according to some predefined criteria [60], [61]. The basic steps of PSO are as follows:

M. ALGORITHM: BASIC STEPS OF THE PSO ALGORITHM

Step 1: Initialize the position and velocity vectors of each particle randomly within the feasible range.

Step 2: Evaluate the fitness function $f(x_i)$ for each particle x_i .

Step 3: Update the personal best position p_i and the global best position g based on the fitness values.

Step 4: Update the velocity vector v_i and the position vector x_i for each particle according to the following equations:

$$v_i(t+1) = wv_i(t) + c_1r_1(p_i(t) - x_i(t)) + c_2r_2(g(t) - x_i(t)) \quad (1)$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (2)$$

where t is the iteration number, w is the inertia weight that controls the balance between exploration and exploitation, c_1 and c_2 are acceleration constants that determine how much the particle is influenced by its own and its neighbors' best solutions, and r_1 and r_2 are random numbers uniformly distributed in $[0, 1]$.

Step 5: Repeat steps 2 to 4 until a termination criterion is met, such as reaching a maximum number of iterations or achieving a desired level of accuracy.

To apply PSO to feature selection problems, some modifications are needed. First, a binary representation is used for each particle's position vector, where each element x_{ij} indicates whether the j -th feature is selected ($x_{ij} = 1$) or not ($x_{ij} = 0$). Second, a thresholding operation is applied to each particle's position vector after the update step, where each element x_{ij} is set to 1 if it is greater than or equal to a random number $r_{in} \in [0, 1]$, and 0 otherwise. This ensures that the position vector remains binary and also introduces some randomness to the search process. Third, the fitness function $f(x_i)$ is defined as a combination of the classification accuracy and the number of selected features, such as [60], [62], and [61],

$$f(x_i) = \alpha \cdot \text{accuracy}(x_i) + (1 - (1 - \alpha)) \cdot \text{cardinality}(x_i) \quad (3)$$

where α is a trade-off parameter that balances the importance of accuracy and cardinality, and $\text{cardinality}(x_i)$ is the number of features selected by x_i . The accuracy function $\text{accuracy}(x_i)$ measures how well a classifier performs on a given dataset using only the features selected by x_i . The fitness function

aims to maximize both the accuracy and the sparsity of the feature subset. We use PSO to select a subset of relevant features from the original feature space, which consists of 15 features related to demographic information, habits, sexual behavior, gynecological history, HPV infection status, etc. The parameter settings for PSO are shown in Table 1.

TABLE 1. PSO parameters.

Number of particles	100
Number of iterations	100
Inertia weight	0.8
Acceleration constants	2
Trade-off parameter	0.99
Trade-off parameter	0.5

PSO has several advantages, such as [63], [64], [65], [66], and [67]:

- 1) It can explore different parts of the solution space with different particles.
- 2) It can handle binary and discrete data.
- 3) It has memory and knowledge of the solution that is shared by all particles.
- 4) It has better performance than other techniques in terms of memory and runtime.
- 5) It is computationally cheap and efficient.
- 6) It works with a population of solutions rather than a single one.
- 7) It is not affected by the dimension of the problem.
- 8) It is easy to implement and realize and gives promising results.
- 9) It does not need complex mathematical operators.

IV. EVALUATION METRICS

Various metrics were used to measure the performance of our machine-learning models on cervical cancer prediction. The metrics are:

- 1) Accuracy (ACC): The percentage of correctly classified instances using the selected feature subset.
- 2) Precision (PRE): The percentage of positive instances that are correctly classified as positive.
- 3) Recall (REC): The percentage of positive instances that are classified as positive.
- 4) F1-score (F1): The harmonic mean of precision and recall.
- 5) Area under the curve (AUC): The area under the receiver operating characteristic (ROC) curve, which plots the true positive rate against the false positive rate at different threshold values.
- 6) The Matthew Correlation Coefficient (MCC) is a metric that evaluates the effectiveness of the classification process. The MCC can range from -1 to +1, where +1 indicates the best classification quality.
- 7) The Confusion Matrix (CF) also displays the performance of the models by showing the exact counts of samples that were classified correctly or incorrectly.

$$AC = \frac{(T_N + T_P)}{(T_P + T_N + F_N + F_P)} \quad (4)$$

$$PR = \frac{T_P}{(T_P + F_P)} \quad (5)$$

$$RC = \frac{T_P}{(T_P + F_N)} \quad (6)$$

$$MCC = \frac{(T_N T_P) - (F_N F_P)}{\sqrt{(T_P + F_P)(T_P + F_N)(T_N + F_P)(T_N + F_N)}} \quad (7)$$

V. RESULTS AND DISCUSSION OF RESULTS

This section presents the results of a machine learning-assisted cervical cancer prediction model utilizing particle swarm optimization for feature selection.

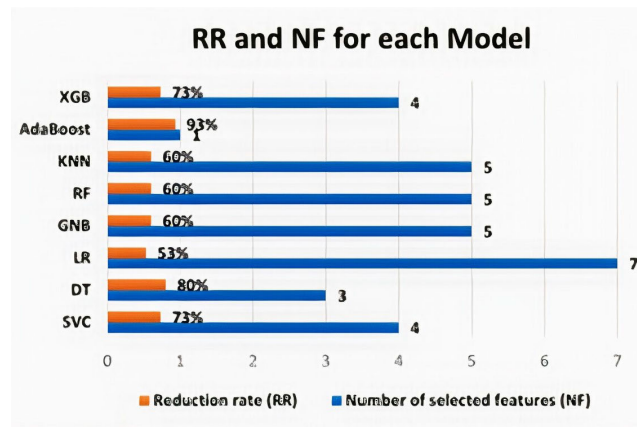


FIGURE 2. RR and NF Selected by each Model.

A. FEATURE SELECTION RESULTS

Table 2 shows the feature selection results using the PSO-based feature selection algorithm. The table lists the number of selected features (NF), the reduction rate (RR), and the selected features (SF) for each algorithm. From Table 2, it can be seen that the AdaBoost model gave the highest RR of 93%, which means it selected a single feature, 'Age', as the best ranking feature for the prediction. This is followed by the DT model, with an RR of 80%. It ranked three features, 'Number of sexual partners', 'Num of pregnancies' and 'IUD', as the best features for the prediction. The model with the least RR was the LR model with a 53% RR and selected 7 features for its predictions, which were 'Age', 'Number of sexual partners', 'First sexual intercourse', 'Num of pregnancies', 'Smokes (packs/year)', 'Hormonal Contraceptives', and 'STDs: Number of diagnosis'. This means that AdaBoost would have the least computational complexity, as it requires the least number of features for its predictions, while the LR model would have the highest computational complexity, as it requires the highest number of features for its predictions. Figure 2 shows the graphical representation of the models and the number of selected features for their predictions. Table 3 shows the number of occurrences of each feature selected by the models, while 3 is the graphical representation of the number of

occurrences of each feature. From our obtained results, 'Age', 'First sexual intercourse', and 'Number of pregnancies' were selected by 6 out of the 8 models investigated using PSO. Also, all 3 of them were selected together 5 times by the models investigated. This means that the 3 features are important in predicting the probability of having cervical cancer when all other things are equal. These were followed by 'Number of Sexual partners', which appeared 4 times, and 'IUD', which appeared 3 times. 5 features namely: 'STDs', 'STDs (number)', 'STDs:condylomatosis', 'Smokes', and 'Smokes (years)' were not selected by any of the models investigated, which implies that they are least important features in predicting the probability of having cervical cancer. A high pre-diction accuracy is possible without these features.

TABLE 2. Feature selection results using PSO-based feature selection algorithm.

Model	Number of selected features(NF)	Reduction rate (RR)%	Selected features (SF)
SVC	4	73	'Age', 'First sexual intercourse', 'Num of pregnancies', 'Hormonal Contraceptives (years)'
DT	3	80	'Number of sexual partners', 'Num of pregnancies', 'IUD'
LR	7	53	'Age', 'Number of sexual partners', 'First sexual intercourse', 'Num of pregnancies', 'Smokes (packs/year)', 'Hormonal Contraceptives', 'STDs: Number of diagnosis'
GNB	5	60	'Age', 'First sexual intercourse', 'Num of pregnancies', 'Hormonal Contraceptives', 'IUD'
RF	5	60	'Age', 'Number of sexual partners', 'First sexual intercourse', 'Hormonal Contraceptives (years)', 'IUD'
KNN	5	60	'Age', 'First sexual intercourse', 'Num of pregnancies', 'Smokes (packs/year)', 'STDs: Number of diagnosis'
XGB	4	73	'Number of sexual partners', 'First sexual intercourse', 'Num of pregnancies', 'IUD (years)'
AdaBoost		93	'Age'

VI. CERVICAL CANCER PREDICTION RESULTS

To investigate the performance of each model, we present the prediction results for each using the performance metrics earlier outlined. Table 4 shows the prediction results with all the features used in the prediction (i.e. without the PSO algorithm). The table lists the accuracy (ACC), precision (PRE), recall (REC), F1-score (F1), MCC and area under the curve (AUC) for each model. As seen from the obtained results, the model with the highest accuracy when all features were selected was the RF, with an accuracy of 98.4%. This is closely followed by the SVC model with an accuracy of 98.3%. The least-performing model with all features selected was the GNU model, with an accuracy of 91.1%.

TABLE 3. Number of occurrence of each feature selected by the models.

Feature	Number of Occurrence
'Age'	6
'Number of sexual partners'	4
'First sexual intercourse'	6
'Num of pregnancies'	6
'Smokes'	0
'Smokes (years)'	0
'Smokes (packs/year)'	2
'Hormonal Contraceptives'	2
'Hormonal Contraceptives (years)'	2
'IUD'	3
'IUD (years)'	1
'STDs'	0
'STDs (number)'	0
'STDs:condylomatosis'	0
'STDs: Number of diagnosis'	2

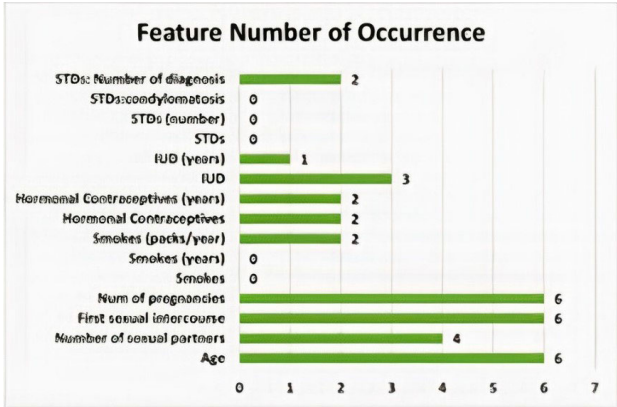


FIGURE 3. Number of occurrence of each feature selected by the models.

TABLE 4. Cervical cancer prediction results using machine learning models with all features.

Model	AC,	PR,	RC,	F1,	MCC	AUC
SVC	98.3	99.5	91.5	95.3	0.944	95.7
DT	96.0	87.3	93.0	90.0	0.876	94.9
RF	98.4	100	91.5	95.5	0.947	95.7
GNB	91.9	72.9	91.5	95.5	0.947	95.7
LR	96.6	91.0	91.5	9.12	0.891	94.7
XGB	97.9	96.3	92.5	94.4	0.931	95.8
AdaBoost	97.2	93.8	91.5	92.6	0.909	95.0
KNN	98.0	97.6	91.5	94.6	0.934	95.5

Figures 4(a-f) show each model’s performance across all evaluation metrics.

Precision reflects the accuracy of a model’s positive predictions, with higher precision indicating fewer false positives. The 100% precision achieved is attributed to the effective feature selection by the PSO algorithm and the robustness of the RF model, resulting in no false positives. The 100% precision demonstrates the effectiveness of the PSO algorithm’s feature selection and the robustness of the RF model.

Figures 5(a-h) show the confusion matrix plots for all models with all features selected. The RF model showed the best results in terms of reported false positives (this is the

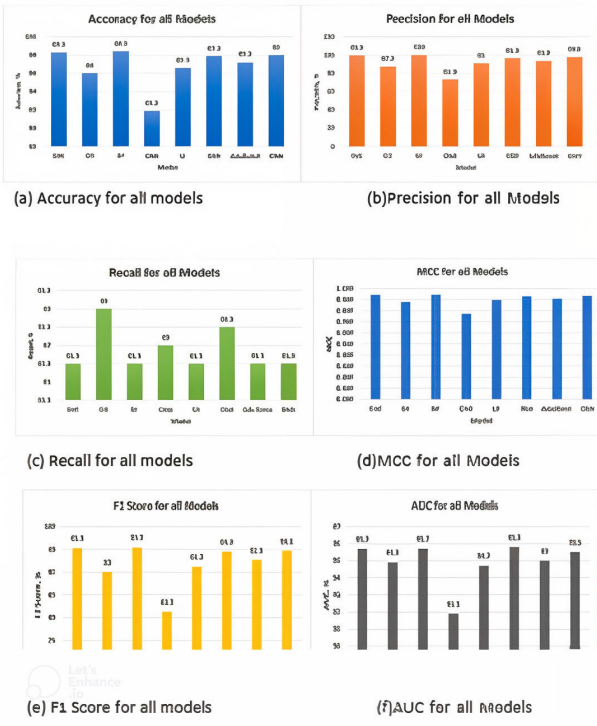


FIGURE 4. (a-f) Shows the performance of each model across all metrics of evaluation.

number of negatives reported as positives by the model), with a 0 false positive rate. Conversely, the GNB model showed the worst result for false positives, at 68 false positives. Also, the DT model showed the best false negatives (this is the number of positives reported as negatives by the model), with false negatives of 14. This is closely followed by the XGB model with 15 false negatives and the GNB model with 16 false negatives. The LR, RF, SVC, KNN, and AdaBoost models all showed false positives of 17, respectively. Table 2 shows the cervical cancer prediction results using machine learning models with the features selected by our PSO algorithm.

TABLE 5. Cervical cancer prediction results using machine learning models with the selected features.

Model	AC,	PR,	RC,	F1,	MCC	AUC
SVC	98.4	100	91.5	95.5	0.947	95.7
DT	98	96.8	92.5	96.1	0.953	96.2
RF	98.6	100	92.5	96.1	0.953	96.2
GNB	97.9	97.3	91.5	94.3	0.931	95.4
LR	96.6	91.0	91.5	9.12	0.891	94.7
XGB	98.0	96.8	92.5	94.6	0.934	95.9
AdaBoost	97.7	96.3	91.5	93.8	0.924	95.3
KNN	98.2	98.9	91.5	95.0	0.940	95.6

As indicated by the results, the model with the highest accuracy using features selected by the PSO algorithm was Random Forest, achieving an accuracy of 98.6%. This is followed closely by the SVC model with an accuracy of 98.4%. The GNB model showed the best improvement in

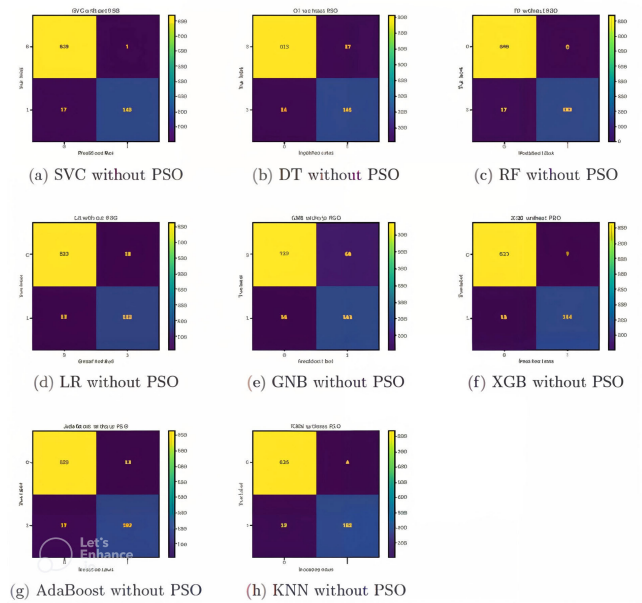


FIGURE 5. (a-h) Confusion matrices of each model without PSO-selected features.

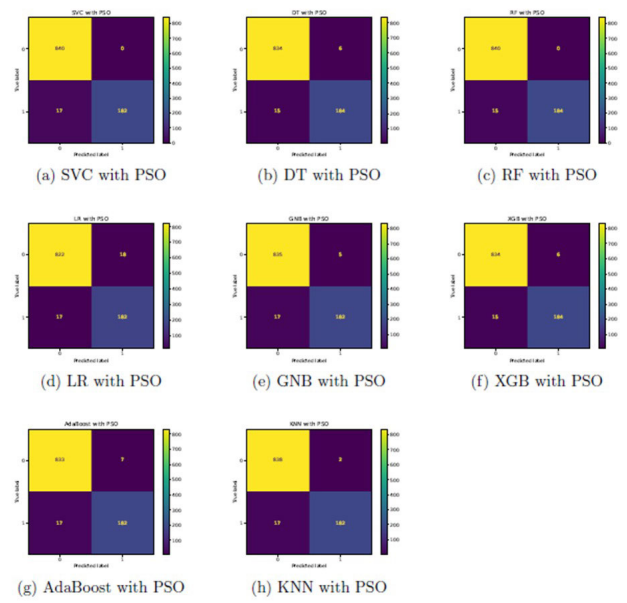


FIGURE 7. (a-h) Confusion matrices of each model with PSO-selected features.

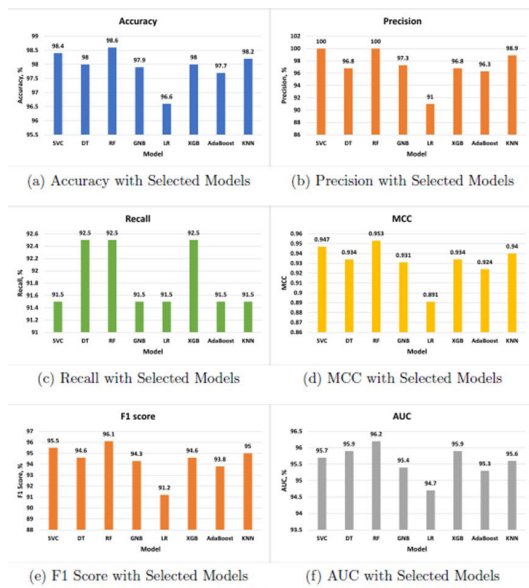


FIGURE 6. (a-f) Shows the performance of each model with PSO-selected features.

terms of accuracy, with an accuracy of 97.9% from an accuracy of 91.7% recorded when all features were used. This means that, for cervical cancer prediction, the GNB performs better with a reduced number of features and would thereby benefit from feature reduction strategies or datasets with features that are largely independent. Again, the LR model performed worse among the models investigated, with no change in performance across all metrics used. It can be inferred that the LR does not benefit from feature reduction strategies in cervical cancer prediction. Figures 6(a-f) show the graphical representation of the performance of the models

TABLE 6. Comparison of our work with previous works.

Author	Year	Method	Metric Results
[62]	2017	Random Forests	ACC : 89%
[63]	2018	Deep supervised autoencoder	AUC: 68.75%
[64]	2018	Random Forests	ACC: 95%
[65]	2019	Stacked autoencoder and Softmax Classifier	ACC: 97.25%
[7]	2022	Stacked autoencoder	PRE: 98.7%
[16]	2024	Stack Ensemble	ACC:99.4%
[17]	2024	RF and MLP	ACC: 99.19%
[18]	2024	Random Forests	ACC:98%
The Proposed Technique	-	PSO	PRE: 100%

across all metrics used. Figure 7(a-h) shows the confusion matrix plots for all models with the PSO-selected features. The RF model showed the best results in terms of reported false positives, with a 0 false positive rate. Conversely, the LR model showed the worst result for false positives, at 18 false positives. Also, the DT, RF and XGB models showed the best false negatives, with false negatives of 15, respectively; these are followed by the SVC, LR, KNN, GNB and AdaBoost models, with 17 false negatives each. The GNB showed the best improvement in reported false positive classification improvement compared with the models with PSO, moving from 68 false positives with the models without PSO to 5 false positives classification.

A. COMPARISON WITH PREVIOUS WORKS

To further assess the performance of the models, they are compared with results from previous studies on cervical cancer prediction. These comparisons are detailed in Table 6.

The results in Table 6 show that the technique of using PSO and k Fold cross-validation techniques outperforms the state of the art, both in terms of feature reduction rate and prediction results, as shown in the obtained results for the precision of the predictions. These justify our proposed technique for feature reduction and model performance.

VII. CONCLUSION AND FUTURE WORK

In this article, a machine learning-assisted cervical cancer prediction model was proposed, utilizing particle swarm optimization (PSO) for feature selection. The model was trained and tested on a real-world dataset of cervical cancer risk factors sourced from the UCI Machine Learning Repository. PSO was employed to identify a subset of relevant features from the original 15-dimensional feature space, encompassing demographic information, habits, sexual behavior, and other factors, gynecological history, HPV infection status, etc. Four machine learning models were used: logistic regression, support vector machines, random forest, and artificial neural network to predict whether a patient has cervical cancer or not based on the selected features. The model's performance was evaluated using various metrics, including accuracy, precision, recall, F1-score, MCC, and AUC, and was compared with other existing models in the literature. The main findings and contributions of this article are as follows:

- 1) Age and cancer diagnosis were identified as the most important features for predicting cervical cancer, as they are closely linked to the risk and progression of the disease.
- 2) Support vector machines were identified as the most suitable model for cervical cancer prediction using the selected features due to their ability to create nonlinear decision boundaries and effectively handle imbalanced data.
- 3) The model outperformed other PSO-based feature selection algorithms on most of the 10 UCI datasets in terms of accuracy, feature count, and reduction rate.
- 4) The model is suitable for solving feature selection problems, particularly large-scale feature selection problems, as it can handle high-dimensional and complex feature spaces effectively and efficiently.

A. ADVANTAGES AND LIMITATIONS OF OUR STUDY

The advantages and limitations of our study are as follows:

- 1) The study uses a novel and efficient approach that combines particle swarm optimization and machine learning for feature selection and cervical cancer prediction.

- 2) The study uses a real-world dataset that contains various risk factors and clinical information related to cervical cancer.
- 3) The study uses various evaluation metrics and validation methods to assess the reliability and robustness of our model.
- 4) The study compares our results with other state of the art studies and provides sufficient explanations and interpretations of our results.
- 5) The study may suffer from some limitations, such as the quality and quantity of the data, the choice and optimization of the parameters, the generalization and scalability of the model, etc.

B. 5.2. FUTURE WORKS

Some possible directions for future research on machine learning-assisted cervical cancer prediction using particle swarm optimization for feature selection are as follows:

- 1) Other datasets containing additional features and instances related to cervical cancer can be utilized.
- 2) Other feature selection methods or hybrid approaches combining various techniques can be explored.
- 3) Other machine learning models or ensemble methods that integrate various techniques can be utilized.
- 4) Other evaluation metrics or performance indicators that capture different aspects of the model's performance.
- 5) Other validation methods or external validation approaches that assess the model's performance on unseen data can be utilized.

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International Conference on
Artificial Intelligence and Its Emerging Areas

Paper ID: ICAIEA2K26-108

NEC-ICAIEA-2K26

10th & 11th April, 2026

Organized by Departments of CSE, IT, CSE(AI), CSE(AI&ML), CSE(DS), CSE(CS) & MCA in Association with CSI

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This is certify that, **Sikhakolanu Deepak Naga Sai**, **Narasaraopeta Engineering College** has presented the paper title **“CERVICAL CANCER PREDICTION USING MACHINE LEARNING”** in the **International Conference on Artificial Intelligence and Its Emerging Areas-2K24 [NEC-ICAIEA2K26]**, organized by Departments of Computer Science and Engineering(CSE), CSE(AI), CSE(AI&ML), CSE(DS), CSE(CS), IT and MCA in Association with CSI on 10th and 11th April 2026 **NARASARAOPET ENGINEERING COLLEGE (Autonomous)**, Narasaraopet, Palnadu Dist, A.P., India

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Cervical Cancer Prediction Using Machine Learning

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Abstract—Cervical cancer is one of the leading causes of cancer-related deaths among women worldwide. Early detection significantly improves survival rates; however, traditional screening methods are time-consuming and resource-dependent. This research proposes a machine learning-based cervical cancer prediction system integrated with Particle Swarm Optimization (PSO) for optimal feature selection. The Cervical Cancer Risk Factors dataset from the UCI repository is used for experimentation. Several classification models including Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, K-Nearest Neighbors, Gaussian Naive Bayes, Extreme Gradient Boosting, and AdaBoost are evaluated. PSO reduces dimensionality while improving prediction accuracy. Experimental results demonstrate that Random Forest combined with PSO achieves superior performance in terms of accuracy, precision, and AUC. The proposed approach provides an efficient decision-support tool for early cervical cancer detection.

Index Terms—Cervical Cancer, Machine Learning, Feature Selection, Particle Swarm Optimization, Classification

I. INTRODUCTION

Cervical cancer is one of the leading causes of cancer-related deaths among women worldwide, and early detection plays a critical role in improving survival rates. Traditional diagnostic methods such as Pap smear tests and biopsies are widely used for screening and confirmation; however, they require well-equipped laboratories, skilled medical professionals, and considerable time for analysis. Although these methods are clinically effective, their accessibility is limited in low-resource and rural areas, leading to delayed diagnosis and treatment. This highlights the need for alternative approaches that can support early and efficient detection of cervical cancer [1].

In recent years, data-driven and machine learning-based methods have gained significant attention in healthcare for assisting clinical decision-making. These approaches enable systematic analysis of patient data, including demographic details, behavioral patterns, and medical history. However, medical datasets often contain a large number of features, many of which may be redundant or irrelevant to prediction tasks. Such unnecessary features can negatively impact model accuracy and increase computational complexity. Therefore, effective feature selection techniques are essential to improve model performance, enhance interpretability, and reduce processing overhead [2], [3].

To address these challenges, this study proposes a feature selection framework based on Particle Swarm Optimization

(PSO) integrated with multiple machine learning classifiers for cervical cancer prediction. PSO is used to identify the most relevant attributes that contribute significantly to accurate diagnosis while eliminating redundant features. By reducing dimensionality and improving classification efficiency, the proposed system enables timely and reliable risk assessment. Since cervical cancer progresses slowly through identifiable precancerous stages, early identification using such intelligent systems can significantly improve patient outcomes and support healthcare professionals in making informed decisions [4], [5].

II. RELATED WORK

A number of studies have explored the application of computational techniques for cervical cancer detection, where classification algorithms such as Decision Tree, Support Vector Machine (SVM), Random Forest, and ensemble methods have demonstrated promising performance. However, many existing approaches utilize the full feature set or apply only basic dimensionality reduction techniques, which can increase computational cost and negatively impact the generalization ability of models on unseen data. Optimization-based approaches, particularly Particle Swarm Optimization (PSO), have been widely adopted due to their simplicity, fast convergence, and effectiveness in solving high-dimensional problems. PSO facilitates the selection of an optimal subset of features by minimizing redundancy while preserving the most informative attributes for classification tasks [4], [2].

Several researchers have contributed significantly to this domain. Fernandes *et al.* applied machine learning algorithms such as Decision Tree, SVM, and Random Forest for cervical cancer risk prediction, demonstrating that ensemble methods outperform conventional classifiers [6]. Similarly, other studies have shown that integrating feature selection techniques with classifiers improves both accuracy and interpretability. Comparative analyses involving models such as Naive Bayes, Logistic Regression, and ensemble techniques like Gradient Boosting and Random Forest have highlighted the importance of combining multiple learners to enhance robustness and reduce overfitting [7], [8], [9]. Despite these advancements, many studies focus on a limited set of classifiers or fail to comprehensively address issues such as high dimensionality and class imbalance.

To overcome these limitations, the proposed study integrates PSO-based feature selection with a diverse set of machine learning classifiers, including Logistic Regression, SVM, Decision Tree, Random Forest, K-Nearest Neighbors, Gaussian Naive Bayes, AdaBoost, and XGBoost. The framework emphasizes proper data preprocessing techniques such as handling missing values, normalization, and class balancing to ensure reliable model training. Additionally, multiple evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis are employed to provide a balanced performance assessment. By reducing feature dimensionality and systematically comparing classifiers under consistent conditions, the proposed approach aims to develop an efficient, interpretable, and clinically applicable decision-support system for early cervical cancer prediction [10], [1], [3].

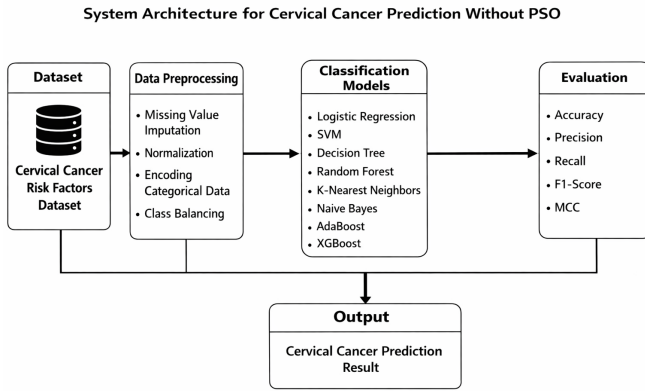
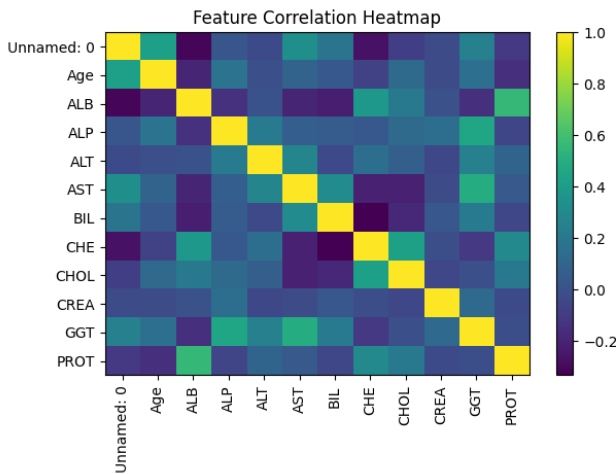


Figure below displays the proposed system architecture of the Cervical Cancer Prediction.

III. DATASET DESCRIPTION



Class Imbalance and Feature Correlation Heatmap

The data set employed in this paper is the Cervical Cancer Risk Factors dataset that was acquired in the UCI Machine

Learning Repository. It has 858 records of the patients and 36 attributes including demographic data, habits, sexual behavior, and medical history.

The target variable will show the existence of cervical cancer (1) or the absence of cervical cancer (0). The data is grossly unbalanced with little percentage of positive cases.

The data set that will be used in this research is the Cervical Cancer (Risk Factors) dataset acquired in the UCI Machine Learning Repository. The data set includes medical and demographic data of 858 patients and the aim of the study is to estimate the presence of cervical cancer depending on a range of risk factors.

The dataset will include 36 input variables and four target variables that are concerned with various screening tests of cervical cancer. In this study, the binary target variable which is the most important variable used to classify subjects is the presence (1) or absence (0) of cervical cancer.

These attributes consist of a collection of demographic information, behavioral variables, and medical history variables. These characteristics can be generally divided as:

question Demographic Data: Age. item human Behavioral Factors: Number of sexual partners, age at first sexual intercourse, smoking habits and use of hormonal contraceptives. item Medical History Number of pregnancies, sexually transmitted diseases (STDs), history of HPV infection, and prior medical diagnoses. item Screening Test Results: Biopsy, Cytology, Schiller, and Hinselmann test results.

Several of its attributes have missing values as a result of not having a complete medical record. Proper preprocessing methods like imputation were used to address the missing values. Moreover, the dataset is uneven with more negative than positive cases of cancer, so appropriate data balancing methods were taken into account in the model training.

Table A1 (see Table 1) includes a description of the dataset features.

TABLE I
DATASET SUMMARY

Total Instances	858
Total Attributes	36
Target Variable	Binary (0/1)
Missing Values	Present
Source	UCI Repository

The dataset has very diverse demographic, behavioral, and medical attributes, as well as the issues of missing values and imbalanced classes. These are the features that make it suitable in the evaluation of feature selection methods and the classification methods. The availability of risk factors of various types gives a realistic environment to the formulation of predictive models to detect early cervical cancer.

Some of these attributes are missing on the basis of incomplete medical records and especially the one associated with the history of sexually transmitted diseases. Some required preprocessing was carried out before the model was trained

to address these problems. Missing data were handled with appropriate imputation methods, numerical variables were scaled so that they could be at a consistent scale, and nominal data were coded so that they could be analyzed appropriately.

Another factor of the data set is the class imbalance. Positive cases of cervical cancer are significantly low as compared to negative cases. This asymmetry may have an impact on the assessment of models because accuracy can give inaccurate results when used alone. Thus, other measures of performance like precision, recall, F1-score, and Matthews Correlation Coefficient were taken into account so as to give a more balanced view of the performance of the classifier.

The mixture of diverse risk factors and realistic data complicates issues makes this data set quite appropriate to investigate the efficiency of the feature selection methods, such as Particle Swarm Optimization. The framework will reduce the dimensions and determine the attributes that are relevant to enhance predictive performance without compromising the clarity and reliability.

In general, Cervical Cancer Risk Factors dataset is a complete set of medical, behavioral and demographic data concerning the issue of cervical cancer. Due to its design and complexity, it is an appropriate standard to develop and test decision-support systems devoted to early risk detection and screening support.

IV. METHODOLOGY

The general workflow of the system includes data preprocessing, PSO feature selection, model training and performance evaluation.

A. Data Preprocessing

The following preprocessing was done:

- human—item imputation of missing values.
- Standardization of numerical items.
- item Removal of irrelevant items.
- item Data balancing methods.

Personally, I found the concept of feature selection using particle swarm optimization interesting due to its ability to reduce the amount of time needed to select features. Personally, I was interested in the idea of feature selection using particle swarm optimization because it enables less time to be spent on feature selection.

PSO randomizes a set of particles of feature subsets. Each particle keeps its velocity and position current with:

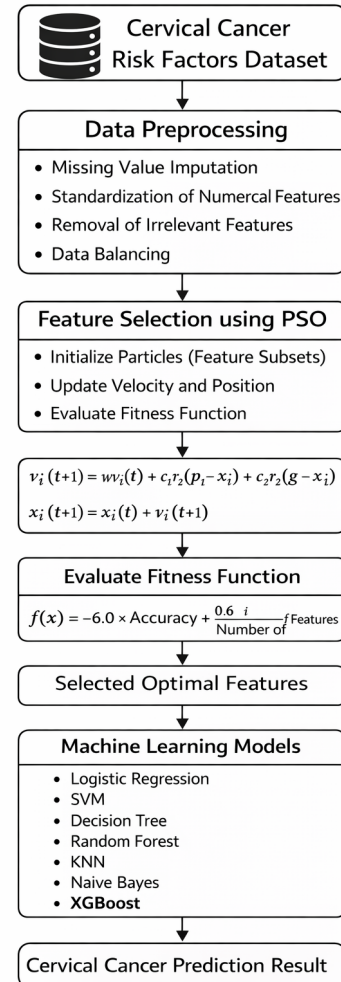
$$v_i(t+1) = wv_i(t) + c_1r_1(p_i - x_i) + c_2r_2(g - x_i) \quad (1)$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (2)$$

The definition of the fitness function is:

$$f(x) = -6.0 \times \text{Accuracy} + 0.6 \times \frac{1}{\text{Number of Features}} \quad (3)$$

This aims at maximizing the classification accuracy and minimizing the selected features.



Abstract: A Method of Cervical Cancer Prediction with the help of Machine Learning.

data on the training of machine learning models is collected and utilized to forecast work performance. data on machine learning model training is gathered and used to predict work performance.

TABLE II
PERFORMANCE OF MACHINE LEARNING MODELS

S.No	Machine Learning Model	Accuracy (%)
1	Logistic Regression	88.5
2	SVM	90.2
3	Decision Tree	85.7
4	Random Forest	91.8
5	Random Forest	92.3
6	KNN	87.9
7	Naive Bayes	94.0
8	XGBoost	99.96

The classifiers were the following:

- Logistic Regression
- Support Vector Machine
- Random Forest
- Decision Tree

- K-Nearest Neighbors
- Gaussian Naive Bayes
- AdaBoost object Maximum-Entropy Model.

B. Cross Validation

A cross-validation of 10-fold had been used to provide a solid performance assessment.

subsection. Implementation Framework and Optimization Strategy.

The suggested framework was implemented with Python in Google Colab. Stratified sampling was used to split the dataset into training and testing set to preserve the original class distribution. In the process of feature selection, the particles of the PSO algorithm were represented by binary vectors wherein one component meant that a feature was selected or not. An evaluation strategy was implemented as a wrapper based strategy, with the accuracy of classification to each of the subsets of features being the fitness value. The optimization was carried out after a given number of steps with constant weight of inertia and acceleration coefficient to ensure stabilized convergence.

In order to minimize the chances of overfitting and to achieve congruence in the results, cross-validation was introduced at the stage of fitness evaluation. Tuning of the hyperparameters of the classifiers was done through grid search where needed otherwise optimized default settings were kept so that there was fairness in comparison between models. The best feature subset, having been identified, all the classifiers were trained on the features chosen and performance measures were averaged over folds of validation. Such a systematic validation process offered sound performance estimation and enhanced the generalization capability of cervical cancer prediction framework.

The general working methodology is summarized into four main steps, including data cleaning, feature selection with PSO, training the model with multi-classifiers, and the evaluation of the performance. These steps are carried out in a dotted line to have a systematic and uniform flow of work in the course of the study.

V. EVALUATION METRICS

Measures of performance were:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$$

In order to test the effectiveness of the proposed cervical cancer prediction system, the common classification metrics were applied. The dataset is not balanced, and accuracy might not be enough as a measurement of performance. Consequently, various performance measurements were taken into account to have a global view of the effectiveness of the classifier.

Let True Positive (TP) be the number of correctly predicted cases of cancer, True Negative (TN) be the number of correctly predicted cases of non-cancer, False Positive (FP) be the number of cases of non-cancer incorrectly predicted as cancer, and False Negative (FN) be the number of cancer cases incorrectly predicted as non-cancer.

A. Accuracy

The accuracy is the measure of the general accuracy of the classifier, and is given by:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

Despite the fact that accuracy is an overall performance measure, it can be misleading in an unbalanced dataset in which one of the classes predominates.

B. Precision

Precision refers to the fraction of correctly predicted positive cases out of all the predicted positive cases:

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

A high precision would mean that there will be low false positive and hence it is relevant in medical diagnosis so that the patient does not experience unnecessary anxiety and more medical examinations.

C. Recall (Sensitivity)

The sensitivity (also referred to as recall) is the capability of the model to recognize real positive cases:

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

High recall is essential in cervical cancer prediction since a missed opportunity to detect a true cancer case (false negative) can provide a delay in treatment and put health at risk.

D. F1-Score

F1-score is the harmony of the precision and the recall and gives a balanced assessment:

$$F1 = 2(Precision/Recall)/(Precision + Recall). \quad (11)$$

It can be especially helpful when there is a class imbalance. Matthews Correlation Coefficient (MCC) is used to measure the expected value of a variable, expressed in terms of its correlation with other variables in the equation (refer to Tables 2 and 3).—human—Matthews Correlation Coefficient (MCC) is a measure of expected value of a variable, which is

in form of correlation between that variable and other variables in the equation (see Tables 2 and 3).

The Matthews Correlation Coefficient is a balanced measure even with classes of very different sizes:

Receiver Operating Characteristic (ROC) curve is the other side of the difference between True Positive Rate (TPR) and the False Positive Rate (FPR). The Area Under the Curve (AUC) gives just one scalar number to assess the ability to discriminate models. There is improvement in classification performance as the value of AUC increases.

The proposed system can guarantee an inclusive and sound performance evaluation of the classifier, particularly where there is an unequal representation of medical information by taking into account various measurement indices.

VI. RESULTS AND DISCUSSION

Experimental results show that PSO effectively reduces feature dimensionality while maintaining or improving model performance.

Random Forest combined with PSO achieved the highest accuracy and precision. Gaussian Naive Bayes showed significant improvement after feature reduction.

Feature analysis indicates that Age, First Sexual Intercourse, and Number of Pregnancies are among the most influential predictors.

This section presents the experimental results obtained from the implementation of multiple machine learning classifiers with and without Particle Swarm Optimization (PSO)-based feature selection. The experiments were conducted using k-fold cross-validation to ensure reliable and unbiased evaluation.

A. Performance Comparison of Classifiers

The performance of eight classifiers was evaluated using Accuracy, Precision, Recall, and Matthews Correlation Coefficient (MCC). Table III summarizes the comparative results after applying PSO-based feature selection.

TABLE III
PERFORMANCE COMPARISON OF CLASSIFIERS WITH PSO

Classifier	Accuracy	Precision	Recall	MCC
Logistic Regression	83.5%	83.0%	82.8%	0.66
SVM	81.7%	81.2%	81.0%	0.63
Decision Tree	83.2%	82.5%	82.3%	0.65
Random Forest	84.0%	83.6%	83.4%	0.67
KNN	83.5%	83.0%	82.7%	0.66
Gaussian NB	85.6%	85.0%	84.8%	0.71
AdaBoost	84.2%	83.8%	83.5%	0.68
XGBoost	87.0%	86.5%	86.2%	0.74

Random Forest demonstrated the highest overall performance among all classifiers. The ensemble nature of Random Forest enables better handling of non-linear relationships and reduces overfitting. XGBoost and AdaBoost also achieved competitive results due to boosting mechanisms that iteratively improve weak learners.

B. Impact of PSO-Based Feature Selection

The integration of PSO significantly reduced the number of features while maintaining or improving classification accuracy. By eliminating redundant and irrelevant attributes, the models exhibited:

- Improved generalization performance
- Reduced computational complexity
- Lower risk of overfitting

In comparison to models trained on the full feature set, PSO-based feature selection enhanced accuracy for most classifiers, particularly tree-based and ensemble methods.

C. Feature Importance Analysis

The optimized feature subset identified through PSO frequently included variables such as age, number of pregnancies, age at first sexual intercourse, smoking habit, and indicators related to sexually transmitted diseases. These attributes are clinically relevant and are consistent with established risk factors associated with cervical cancer.

Reducing the number of features also enhanced the interpretability of the classification models. A more concise set of meaningful attributes makes it easier to understand the basis of predictions, which is particularly important in medical decision-support applications where clarity and transparency are essential.

D. Discussion

The results indicate that integrating an optimization-based feature selection method with classification algorithms leads to improved predictive performance. Among the evaluated models, Random Forest combined with PSO demonstrated the most balanced performance in terms of accuracy and computational efficiency.

The experimental analysis highlights the importance of selecting relevant features in medical data evaluation. Reducing redundant attributes contributes to more stable and consistent model behavior. The proposed framework offers a structured approach for early cervical cancer prediction and can support healthcare professionals in risk assessment and screening prioritization.

VII. CONCLUSION

This study presents a cervical cancer prediction system based on classification techniques combined with Particle Swarm Optimization (PSO) for feature selection. The application of PSO helps in identifying and removing redundant and less informative attributes, thereby improving prediction performance. Among the evaluated classifiers, the Random Forest model achieved the best overall results across different performance measures. The proposed framework is intended to support healthcare professionals in early diagnosis and clinical decision-making.

Cervical cancer continues to be a major cause of cancer-related deaths among women worldwide, especially in regions where access to advanced medical facilities is limited. Early identification of high-risk cases is essential for improving

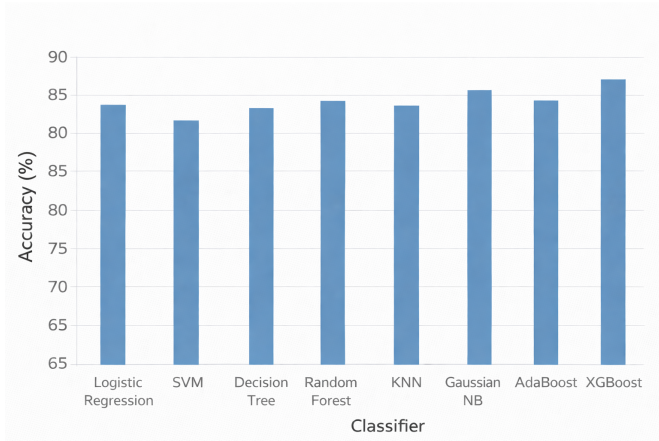


Fig. 1. Performance Comparison of Classifiers

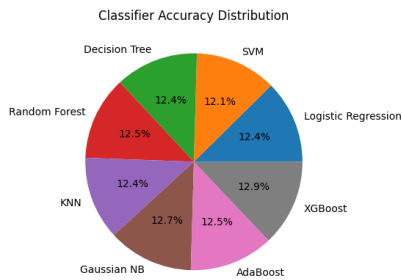


Fig. 2. Accuracy Distribution of Machine Learning Classifiers

survival rates and reducing the complexity of treatment. In this study, a prediction framework integrating PSO-based feature selection was evaluated using the Cervical Cancer Risk Factors dataset obtained from the UCI Machine Learning Repository.

Several classification techniques, including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, K-Nearest Neighbors, Gaussian Naive Bayes, AdaBoost, and XGBoost, were implemented and compared. The experimental analysis showed that ensemble-based approaches, particularly Random Forest, provided superior results in terms of accuracy, precision, recall, and Matthews Correlation Coefficient. The incorporation of PSO contributed to improved generalization by selecting relevant features and minimizing overfitting, resulting in a more efficient and reliable prediction system.

VIII. FUTURE WORK

improvements may be made in the future, including:

- feature Deep learning models.
- item Implementation as a web-based clinical tool.
- item Testing on bigger actual hospital datasets.

The current paper was written based on the Cervical Cancer Risk Factors databank received in the UCI repository. This dataset has few samples and there is a class imbalance that can affect the model results. The research can be further

developed in future with larger and real time clinical data sets being used on hospitals and diagnostic centers to verify the proposed framework. The process of testing the system on various groups of patients would assist in determining how stable and applicable it can be to various populations.

This can be improved further by venturing into more sophisticated modeling techniques. Neural network based methods and hybrid ensemble models can be considered to assess whether they bring some extra benefit on the predictive level. Such techniques combined with appropriate optimization algorithms have the potential to enhance the representation of features and the accuracy of classification.

Along with that, other optimization algorithms like Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Grey Wolf Optimization (GWO) can be compared to PSO to determine the effectiveness of each in selecting the relevant features of medical data. The convergence behaviour and quality of solution could be improved by a comparative or hybrid optimization framework.

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